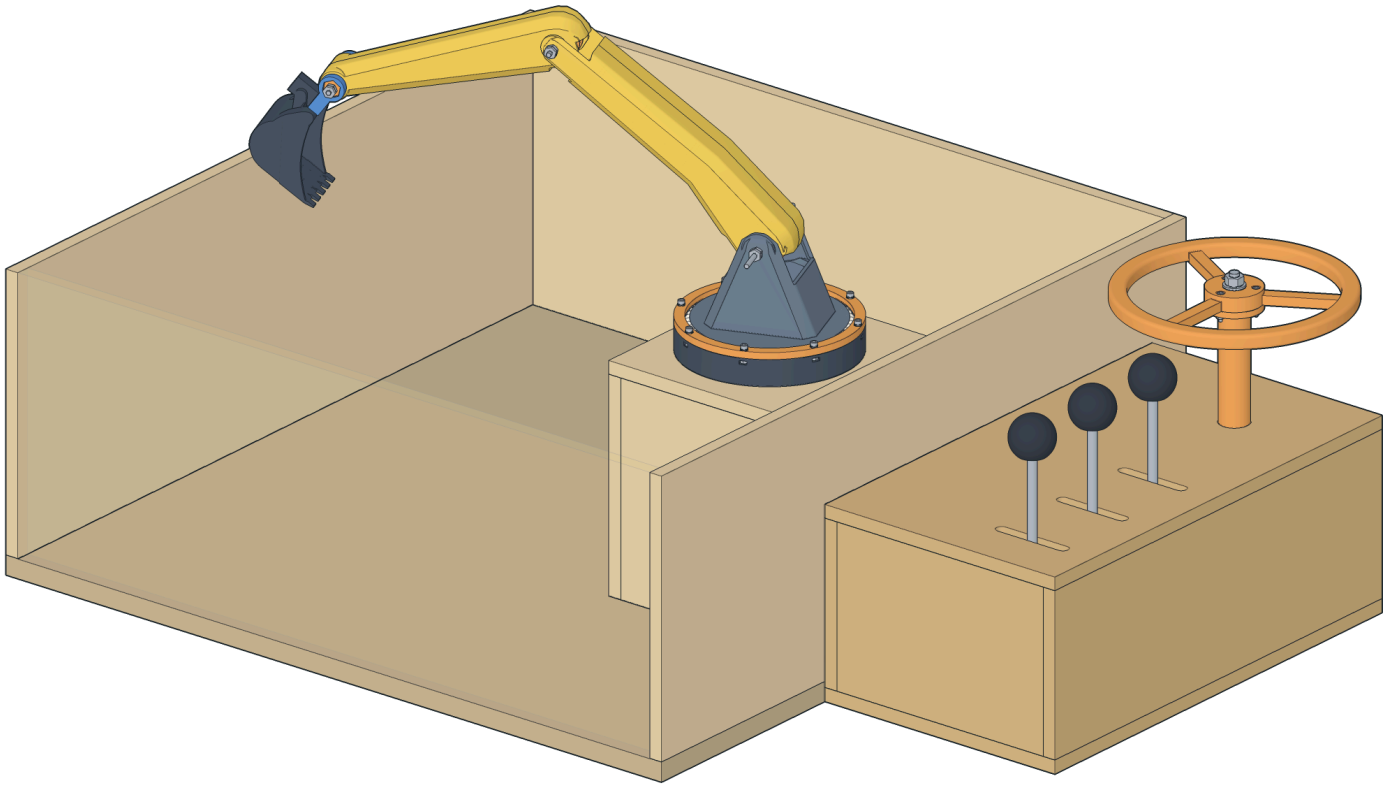
ASSEMBLY & CONFIGURATION MANUAL · MINIMAL BUILD

EX-MA CABLE EXCAVATOR

Minimal design · no PVC conduit · 7/16″ OSB + 5/8″ particle board



Edition 2026-10-11 · minimal design, for the project files in `2026-09-24-EX-MA-simplified/`
Cut sizes in inches for 3 × 7/16″ OSB 2 × 4 ft + 1 × 5/8″ particle board 2 × 2 ft · illustrations rendered from the minimal CAD

26 printed part types · 29 prints
23 wood pieces · 4 rope circuits · no electronics

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Page numbers count the cover as page 1. Figures are numbered in order through the manual.

BEFORE YOU BEGIN

ABOUT THIS MANUAL

This manual builds the **minimal** EX-MA: the same arm, turret, levers and rope circuits as the full design, but with no 2″ PVC conduit. The pedestal stands right against the sandbox wall on two OSB spacer frames, and four 8-32 rods clamp the box, the wall, the spacers and the pedestal together. Everything is sized for the boards you have. Work through the stages in order.

HOW THE PAGES WORK

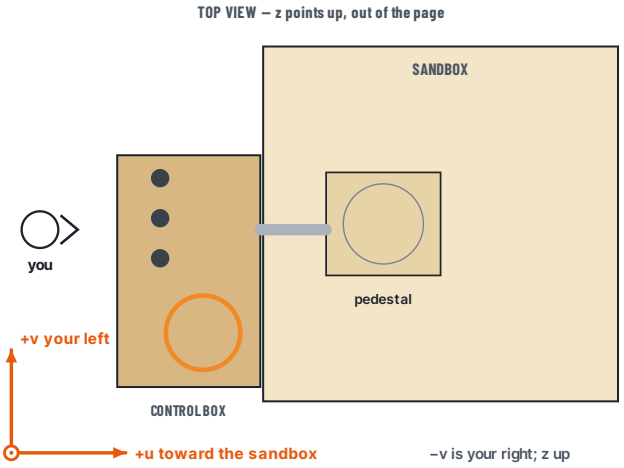
- **Stage header**: the stage number and what it achieves.
- **You need**: every part and fastener for that stage, with its ID (P05 printed, W01 wood, H03 hardware, S04 stock) and quantity. Lay them out first.
- **Numbered steps**: do them in order; each step is one action.
- **Figures**: numbered balloons (1) match the list under the figure. **Orange arrows** show where a part moves to.
- **Check** boxes: stop and verify before going on.

UNITS

Everything you cut or drill is given in inches, to the nearest 1/16″: wood, rods, rope, PTFE tube, PEX and copper, and every hole position. Small grey millimetres follow where the CAD value helps. Printed-part details (M3 screws, Ø60 drums, “3 mm below”) stay metric, as in the project files.

DIRECTIONS USED EVERYWHERE

- **+u** points away from you at the control box, toward the sandbox (“forward”).
- **+v** points to your left; **–v** to your right. **z** points up.



VERIFIED, DERIVED OR ASSUMED?

Tag	Meaning
VERIFIED	Read from the minimal CAD model, its build scripts or the project reports.
DERIVED	Worked out from the model geometry for this manual; the reasoning is given next to it.
ASSUMPTION	Not in the project files: general practice or a choice you can change.
OPEN ITEM	A conflict or gap in the design files. Appendix C lists them with options.

SYMBOLS

WARNING Risk of injury. Read before doing the step.

CAUTION Risk of damaging a part or having to take things apart again.

COMMON MISTAKE Something that is easy to get wrong at this point.

CHECK A verification point. Do not continue until it passes.

TIP Practical advice. Helpful, but not taken from the CAD model.

OPEN ITEM A question found in the design files that you must decide on.

Colours in the illustrations code the part type: printed PLA+ arm shell (PETG) bucket ears original EX-MA dark parts OSB spacer frames particle board metal. Ropes: boom stick bucket slew. Parts fitted in earlier steps are light grey.

READ FIRST

SAFETY AND GOOD PRACTICE

The EX-MA is hand-powered: no motors, springs or electrics. The hazards are the tools you build it with, cut wire rope, small parts, and pinch points between moving printed parts.

CUT WIRE ROPE IS SHARP. Each cut end of 7×19 rope splays into fine, stiff wires that puncture skin. Wrap tape round the rope before cutting and cut through the tape; wear gloves and eye protection.

SMALL PARTS. The 6 mm BBs, M3 and 8-32 nuts and crimp sleeves are choking hazards. Keep the build area and the finished model away from small children.

PINCH POINTS. The slewing ring, the lever hubs and every joint drum can trap fingers while ropes are tensioned. Keep fingers out of the box and the pedestal while someone moves a lever or the wheel.

SAWS, JIGSAWS AND DRILLS. Clamp OSB and particle board before cutting or drilling; support both sides of a cut so the blade does not bind. OSB and particle-board dust is fine wood dust with resin: cut outdoors or with dust extraction, and wear a dust mask.

ADHESIVES. CA glue bonds skin in seconds. Use it with ventilation. Follow the maker's cure times for wood glue, CA, epoxy and silicone; the spacer seal needs its silicone cured before Stage 7.

HEAT. Printed PLA+ softens well below the temperature of a car interior in summer. Do not leave the model in direct sun or in a hot vehicle. ASSUMPTION

DO NOT FORCE A JOINT. Every lever has a hard stop at the end of its slot in the box top. If a lever or the wheel resists before reaching a stop, find the cause (page "Troubleshooting") instead of pushing harder.

SAND AND PARTICLE BOARD. Particle board swells and crumbles if it gets wet. Seal the floor (Stage 3), keep the sand dry and keep it below the pedestal top. ASSUMPTION

WORKSPACE

- A flat table about 4 × 2-1/2 ft; good light (rope holes are Ø2.4 mm).
- Saw horses or a bench for breaking down the 2 × 4 ft OSB panels; a straight edge or a fence for the rip cuts.
- Small trays or bags for hardware: one per stage. Masking tape and a fine marker for labelling every piece.

BEFORE YOU START

- ☐ Read the whole manual once, including Appendix C (open items). Decide items 1–5 before the stage they affect.
- ☐ Check your 5/16" rod: threaded or smooth (Open item 4), and that you have about 3-1/8 ft of it (Stage 2).
- ☐ Check the hardware against Appendix A and the actual-size drawings in "Parts identification".
- ☐ Have at least 25 ft of rope, a 5 m (16 ft) roll of PTFE tube, 57 BBs plus spares, the crimp sleeves and 22 8-32 nuts.
- ☐ Your printed parts are used as printed: old lever hubs, old slew spool, elbow support (its base plate is sawn off in Stage 4).

SYSTEM OVERVIEW

WHAT YOU ARE BUILDING

Three levers and a wheel on a wooden control box drive the EX-MA arm through wire ropes. Each control moves exactly one motion: **boom**, **stick**, **bucket** or **slew**. The ropes leave the box through a printed fitting, cross a short chamber inside the sandbox wall and two spacer frames, enter the pedestal and run up the turret's centre tube and along the arm.

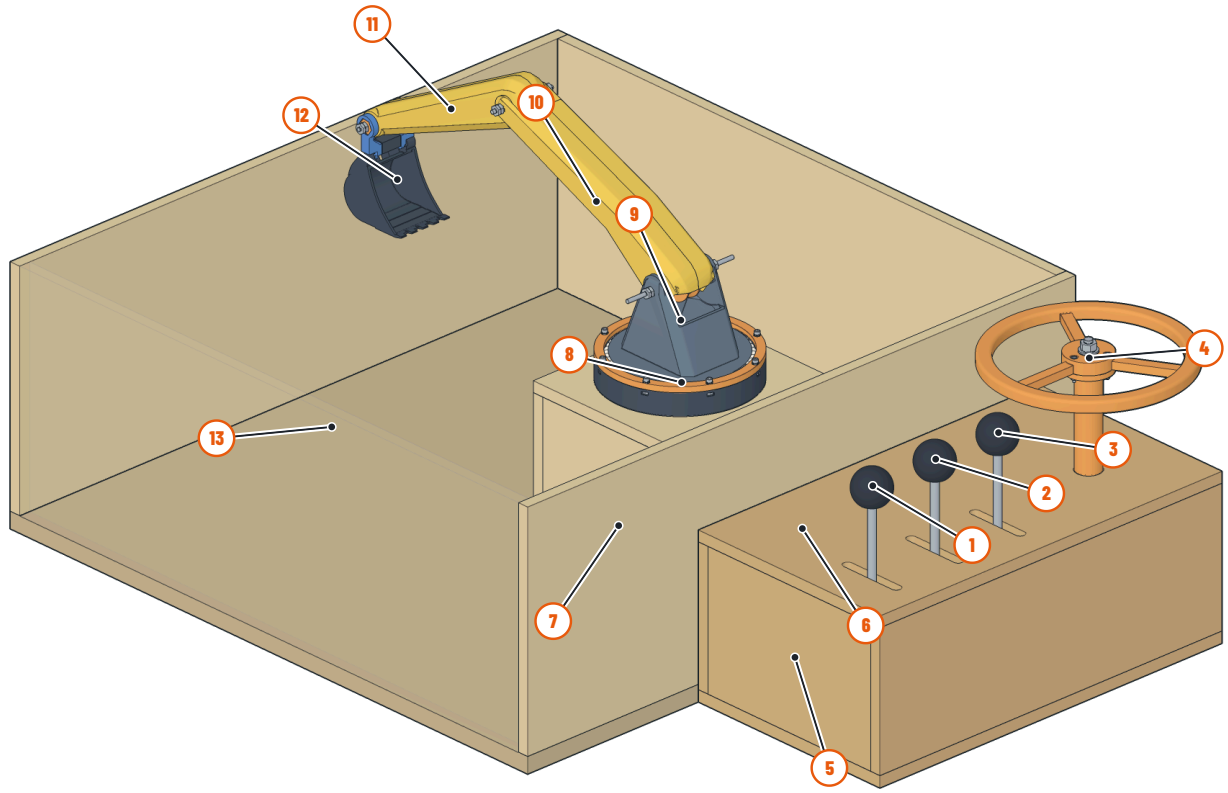


Fig. 1 The complete minimal machine as modelled (joints at their modelled pose, the near sandbox wall see-through).

1. Boom lever
2. Stick lever
3. Bucket lever
4. Slew wheel
5. Control box (OSB)
6. Removable box top: the lever slots are the end stops
7. Pedestal, against the sandbox wall on two spacer frames
8. Base with the printed slewing ring (57 BBs)
9. Tower (turret)
10. Boom
11. Stick
12. Bucket
13. Sandbox, 24″ × 24″

MAIN DIMENSIONS VERIFIED

Sandbox (outside, floor included)	24″ × 24″ × 9-1/2″
Control box (outside, deep × wide × high)	8-7/16″ × 15″ × 6-1/8″
Pedestal (outside)	7-3/8″ × 7-1/16″ × 7-5/16″
Overall length, box rear to sandbox far wall	32-7/16″
Knob tops / slew wheel top above the table	9-3/8″ / 10-3/8″
Base / slewing ring; boom / stick pin to pin	Ø140 mm; 209 / 175 mm

Box and sandbox both stand on the table; no shim. The box is 15″ wide because the lever hubs and slew spool are kept as printed (Appendix D).

SYSTEM OVERVIEW

HOW IT WORKS

Every circuit is a **pull-pull loop**: one rope, anchored on two drums. Turning the control drum pulls one strand and lets out the other by the same amount, so the joint drum, and the joint, turns. Nothing else is connected to that rope, so each control moves only its own joint.

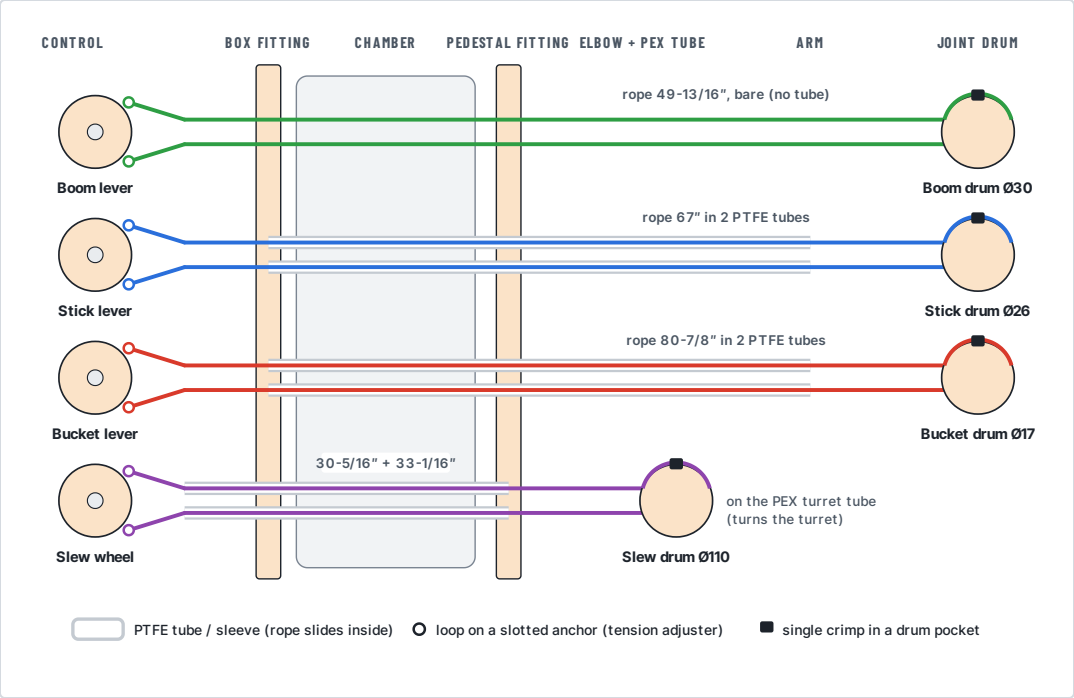


Fig. 2 Rope circuits (the "wiring diagram" of this machine). Each lever drives a Ø60 drum on the common lever axle; the slew wheel drives a Ø110 two-lane spool. Lengths are the cut lengths for the minimal build.

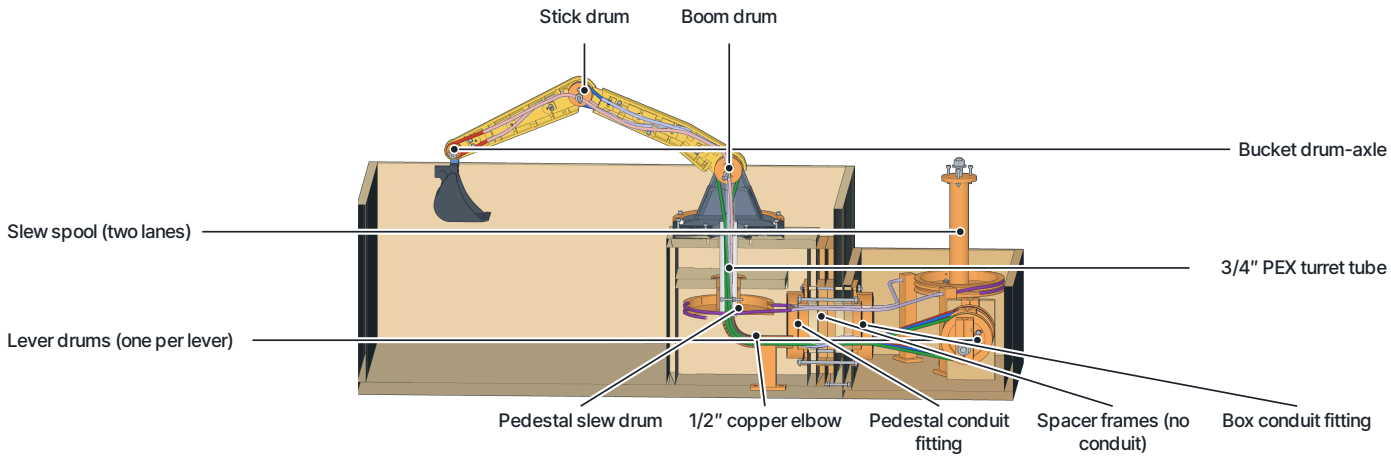


Fig. 3 Every rope as modelled, from its control to its joint (cut through the middle; built pose).

The spacer chamber replaces the conduit

Between the box fitting and the pedestal fitting the ropes and tubes cross a 3-1/8" chamber: the holes in the box front, the sandbox wall and the pedestal wall, and the windows in the two spacer frames, sealed on all sides. The stick and bucket tubes keep their spare length there as a gentle bow (radius about 1-3/16", more than the 5/8" minimum). DERIVED

Why the stick and bucket ropes run in PTFE tubes

Their ropes cross the boom joint (the bucket ropes the stick joint too). Inside a tube anchored at the box fitting and at a stop in the arm, the rope length does not change when another joint moves. The tubes cross each joint through the pivot axis (the pins are split in two for this). The slew drum sits in the pedestal on the 3/4" PEX turret tube. VERIFIED

SYSTEM OVERVIEW

CONTROLS AND SPECIFICATIONS

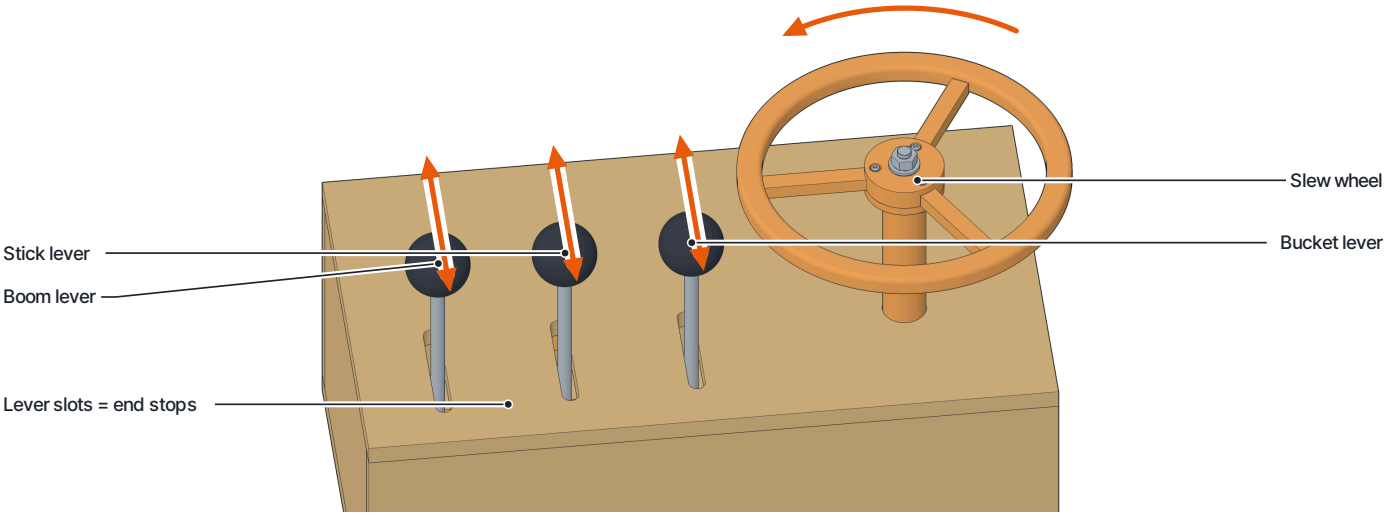


Fig. 4 The operator's view: three levers in slots, the slew wheel on the right. Orange arrows show the lever travel and the wheel.

LEVER AND WHEEL DIRECTIONS

Control	Push forward	Pull back	Joint range	Lever swing
Boom lever	boom down	boom up	-40° … +45°	42.5°
Stick lever	stick out	stick in	-45° … +50°	41.2°
Bucket lever	bucket dump	bucket curl	-55° … +85°	39.7°
Slew wheel	turn left (anticlockwise from above) → arm swings to your left; 1 : 1		−90° … +90°	180°

"Forward" = toward the sandbox (+u). Directions and ranges VERIFIED: architecture notes and kinematics report. You set the lever direction when you rig each rope (Stage 9); a reversed lever is fixed by reversing that rope's wrap on its joint drum.

KEY SPECIFICATIONS VERIFIED

Rope	1/16" galvanized 7×19
Joint drums (pitch Ø)	boom 30 · stick 26 · bucket 17
Control drums	levers Ø60 · slew spool Ø110
Slew drums	Ø110, two lanes
Slewing ring	57 × 6 mm BBs on Ø118
Joint pins	8-32 threaded rod
Axles / handles	5/16" rod
Printed parts	26 types, 29 prints, ≈1.03 kg
Materials	arm shell PETG, rest PLA+

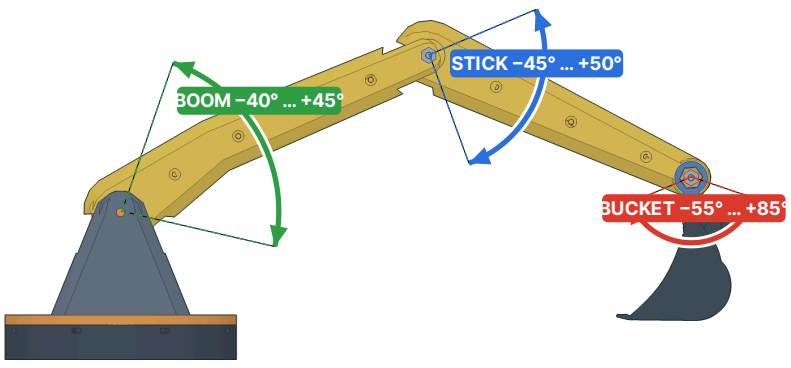


Fig. 5 What each lever moves: the working ranges set by the slots (detail in Stage 12).

PLANNING

BUILD AT A GLANCE

Twelve stages. Stages 1 and 2 are preparation; 3–8 build the machine; 9–12 rig, adjust and test it.

01 p. 14
PRINT & CHECK
29 prints on 7 plates (already printed: check them)

02 p. 15
CUT, DRILL & LABEL
OSB boards, rods, rope, PTFE; label every piece

03 p. 22
SANDBOX
particle-board floor + 4 OSB walls

04 p. 23
PEDESTAL
walls, spacers, fitting + rods, elbow, shelf

05 p. 26
SLEWING RING & TURRET
base, PEX tube, BBs, turret, ring, slew drum

06 p. 28
CONTROL BOX
carcass, lever axle + hubs, spool, posts

07 p. 31
TUBES & BOX MOUNTING
4 PTFE tubes, box onto the rods

08 p. 33
ARM
drums, ropes at the joint end, halves, pins, bucket

09 p. 37
ARM ROPES AT THE LEVERS
zero pose, loops on anchors, tension

10 p. 39
SLEW ROPES
sleeves, drum pockets, spool anchors

11 p. 41
TOP, HANDLES, WHEEL
top panel, handles, knobs, wheel

12 p. 42
TEST & CONFIGURE
ranges, coupling, tension, friction

WHAT IS DIFFERENT IN THE MINIMAL BUILD DERIVED

- **No conduit.** The pedestal stands against the sandbox wall on two glued-on OSB spacer frames (W22, W23). Four 8-32 rods run from the box fitting (P18) to the pedestal fitting (P17) and clamp the whole stack; they replace the 8 M3 fitting bolts.
- **Your boards.** 7/16″ OSB for every panel except the sandbox floor, which is the 5/8″ particle board used whole. With no shim, the sandbox and pedestal walls are cut 5/32″ shorter so the turret and fittings stay at their design heights, and the elbow support loses its base plate.
- **Compact box with your prints.** 8-7/16″ deep, 6-1/8″ high, 15″ wide (old hubs at 2-3/4″ spacing, old spool); handles 5-11/16″; the slew wheel sits 3-3/4″ above the lid.
- **Order.** The rods go into the pedestal in Stage 4; the tubes are seated in P18 and the box goes onto the rods in Stage 7. The lever axle goes in before the left box side; the joint end of each arm rope is rigged while its arm half is open.

OPEN ITEMS TO DECIDE BEFORE YOU START OPEN ITEM

#	Item	Affects
1	Stick-drum key screws: two M3 × 12 enter each through-hole from opposite sides and their tips overlap by 1.6 mm.	Stage 8
2	Stick joining lugs 1 and 2 at the stick root are 2.9 mm apart: their holes merge, so only one screw fits there.	Stage 8
3	Joining-lug screws cannot be reached once a boom or stick is closed.	Stage 8
4	5/16″ nuts on the axles need threaded rod ends; one design note calls the rod “smooth”.	Stages 2, 6, 11
5	Pedestal-to-floor screws are not counted in the bill of materials.	Stage 4
6	Rope tension: the checklist says slide the anchors “outward”; the geometry says inward.	Stages 9, 10
7	Bucket knob to slew-wheel rim clearance is only 5/8″ with the kept spool.	Stage 11

PLANNING

TOOLS AND SUPPLIES

From the project checklist VERIFIED, with the wood tools for your boards. Everything else is in the bill of materials (Appendix A).

HAND TOOLS

2.5 mm hex key	M3 socket head cap screws
2 mm hex key	M3 button head screws (rope anchors)
5.5 mm wrench or nut driver	M3 nuts
11/32″ wrench (×2)	8-32 nuts: jam pairs and through-rods
1/2″ wrench	5/16″ nuts on the axles
Screwdriver (Phillips)	#6 and #8 wood screws
Crimp tool for 1/16″ sleeves	single and double crimp sleeves
Wire-rope cutter	1/16″ 7×19 rope
PTFE tube cutter or new razor blade	cut square
Hacksaw + metal file	5/16″ rod, 8-32 rod, copper; deburr every end
Tape measure, square, straight edge	marking out the boards (1/16″ marks)

POWER TOOLS AND CONSUMABLES

Circular saw (or table saw) + fence	rip and crosscut the OSB; 1/8″ kerf assumed
Jigsaw	the two spacer windows; slot ends
Drill + bits	2-7/16″ hole saw; 1-1/4″ and 1″ spade or Forstner; 7/8″ Forstner; 3/8″, 11/32″, 3/16″, 1/8″, 3/32″
Wood glue	every panel joint; spacers to the pedestal wall
CA or epoxy	bucket ears; knobs; elbow-support post
Silicone	spacer seal against the sandbox wall
Sealer (paint, poly or thinned glue)	particle-board floor, all faces and edges ASSUMPTION
Cable tie (1)	copper elbow to its support
Masking tape + fine marker	label every piece

HARDWARE BAGS BY STAGE VERIFIED

Stage	Fasteners and stock to bag
3 Sandbox	32 × #6 × 1-1/4″
4 Pedestal	41 × #6 × 1-1/4″; 3 × #6 × 3/4″; 4 × 8-32 rod 3-1/16″ + 8 × 8-32 nut; copper elbow; cable tie; glue; silicone
5 Slewing ring	4 × #8 × 1″; 8 × M3 × 12; 2 × M3 × 25; 10 split-lock washers; 10 M3 nuts; 57 BBs; PEX tube
6 Control box	32 × #6 × 1-1/4″; 11 × #6 × 3/4″; 5/16″ rods 8-1/16″ + 10-13/16″; 3 × 5/16″ nut + washer
7 Tubes + box	4 arm PTFE tubes; 2 × M3 × 30; 4 flat washers; 2 M3 nuts (the rod nuts are in the Stage 4 bag)
8 Arm	8 × M3 × 12 (or M3 × 8, Open item 1); 8 × M3 × 20 + 8 nuts; 8-32 rods 2 × 1-5/8″, 2 × 7/8″, 1 × 2-9/16″; 12 × 8-32 nuts; 3 single crimps
9 Arm ropes	6 × M3 × 10 button head; 12 flat washers; 6 split-lock washers; 6 nuts; 6 double crimps
10 Slew ropes	2 × M3 × 10 button head; 4 flat washers; 2 split-lock washers; 2 nuts; 2 single + 2 double crimps; 2 sleeves
11 Top	16 × #6 × 1-1/4″; 9 × M3 × 20; 9 split-lock washers; 9 nuts; 3 handle rods 5-11/16″; 1 × 5/16″ nut + washer

PARTS IDENTIFICATION

HARDWARE AT ACTUAL SIZE

Print this page at 100 % (no “fit to page”) and lay each screw or rod on its drawing to identify it. The scale bar at the bottom must measure 4″. VERIFIED

M3 SCREWS — HEAD UP, ACTUAL SIZE



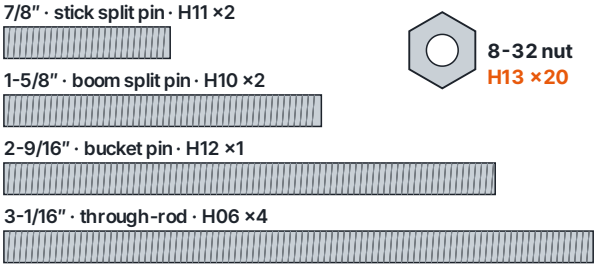
M3 × 10	M3 × 12	M3 × 20	M3 × 25	M3 × 30
button head	socket cap	socket cap	socket cap	socket cap
H01 · ×8	H02 · ×16	H03 · ×17	H04 · ×2	H05 · ×2

M3 NUT AND WASHERS



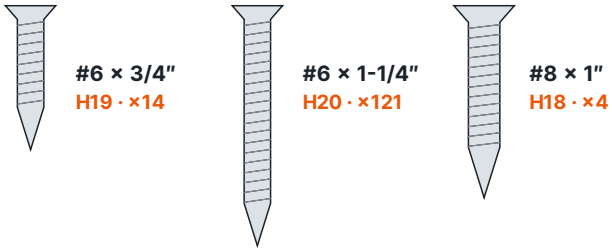
M3 nut	flat washer	split-lock
H07 · ×37	H08 · ×20	H09 · ×27

8-32 ROD AND NUT (HILLMAN)



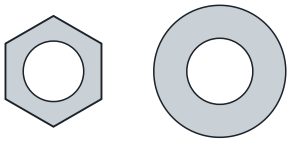
8-32 nut
H13 ×20

WOOD SCREWS (FLAT HEAD)



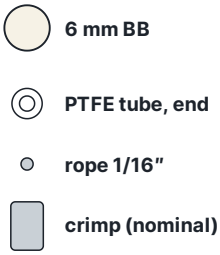
#6 × 3/4″	#6 × 1-1/4″	#8 × 1″
H19 · ×14	H20 · ×121	H18 · ×4

5/16″ NUT AND WASHER



H17 · ×4 each (axle ends)

STOCK (ACTUAL SIZE)



- 6 mm BB
- PTFE tube, end
- rope 1/16″
- crimp (nominal)

4 inches: measure this to check the print is at 100 %

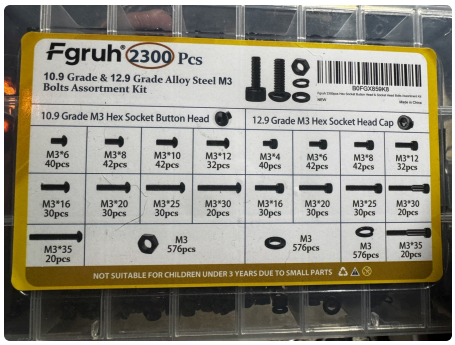


Photo Your M3 kit (Fgruh, 2300 pc): socket head caps and button heads in 4–35 mm, nuts, flat and split-lock washers.



Photo Your Hillman parts: 3 ft of 8-32 threaded rod and 22 hex nuts. The minimal build uses about 20″ of the rod and 20 nuts.

M3 × 12 VS M3 × 20. They look alike in the bag. Lay them on the drawing above before each step.

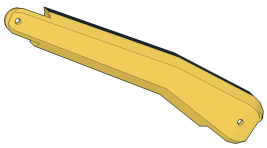
WOOD SCREWS are your own stock. Flat-head #6 and #8 are drawn; pan heads will not sit flush in the printed counterbores. ASSUMPTION

PARTS IDENTIFICATION

PRINTED PARTS

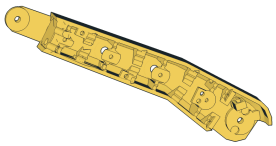
26 part types, 29 prints. Each card shows the part ID used in this manual, the quantity, the material and slicer plate, and the STL file. VERIFIED

P01 ×1



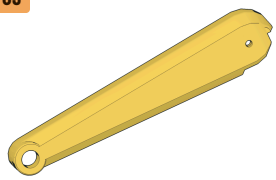
Boom half, left (+v)
PETG · plate arm 1
STL/2026-09-24-boom-half-left.stl

P02 ×1



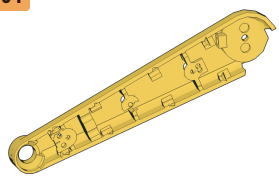
Boom half, right (−v)
PETG · plate arm 1
STL/2026-09-24-boom-half-right.stl

P03 ×1



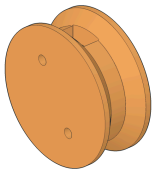
Stick half, left (+v)
PETG · plate arm 1
STL/2026-09-24-stick-half-left.stl

P04 ×1



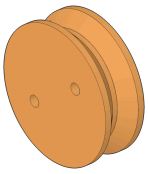
Stick half, right (−v)
PETG · plate arm 1
STL/2026-09-24-stick-half-right.stl

P05 ×1



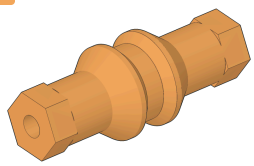
Boom drum (Ø30 pitch)
PLA+ · plate other 6
STL/2026-09-24-boom-drum.stl

P06 ×1



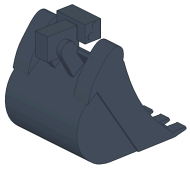
Stick drum (Ø26 pitch)
PLA+ · plate other 6
STL/2026-09-24-stick-drum.stl

P07 ×1



Bucket drum-axle (Ø17 pitch)
PLA+ · plate other 6
STL/2026-09-24-bucket-drum-axle.stl

P08 ×1



Bucket
PLA+ · plate other 6
STL/2026-09-24-bucket.stl

P09 ×1



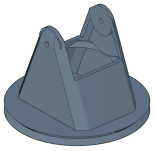
Bucket ear, left (G)
PLA+ · plate other 6 · new file
STL/2026-09-29-bucket-ear-left-cad.stl

P10 ×1



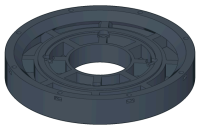
Bucket ear, right (H)
PLA+ · plate other 6 · new file
STL/2026-09-29-bucket-ear-right-cad.stl

P11 ×1



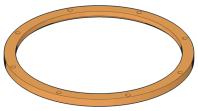
Tower (turret)
PLA+ · plate other 4
STL/2026-09-24-tower.stl

P12 ×1



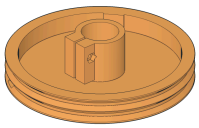
Base (lower race)
PLA+ · plate other 4 · new file
STL/2026-09-29-base.stl

P13 ×1



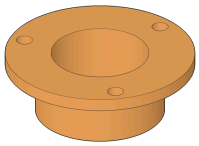
Slewing-ring retaining ring
PLA+ · plate other 5
STL/2026-09-24-slewing-ring-retaining-ring.stl

P14 ×1



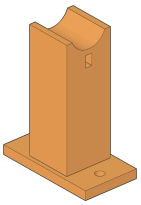
Pedestal slew drum (Ø110)
PLA+ · plate other 1
STL/2026-09-24-slew-drum.stl

P15 ×1



PEX bushing
PLA+ · plate other 1
STL/2026-09-24-slew-tube-bushing.stl

P16 ×1

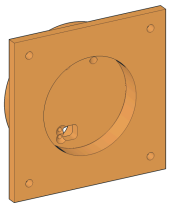


Elbow support (base plate sawn off, Stage 4)
PLA+ · plate other 1
STL/2026-09-24-elbow-support.stl

PARTS IDENTIFICATION

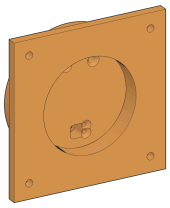
PRINTED PARTS (CONTINUED)

P17 ×1



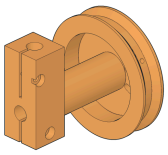
Conduit fitting, pedestal end
PLA+ · plate other 1
`STL/2026-09-24-conduit-end-fitting-pedestal.stl`

P18 ×1



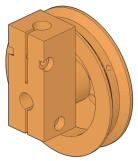
Conduit fitting, box end
PLA+ · plate other 3
`STL/2026-09-24-conduit-end-fitting-box.stl`

P19 ×1



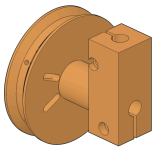
Lever hub, boom
PLA+ · plate other 2
`STL/2026-09-24-lever-hub-boom.stl`

P20 ×1



Lever hub, stick
PLA+ · plate other 2
`STL/2026-09-24-lever-hub-stick.stl`

P21 ×1



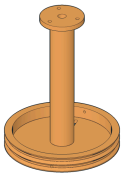
Lever hub, bucket
PLA+ · plate other 3
`STL/2026-09-24-lever-hub-bucket.stl`

P22 ×3



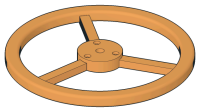
Lever knob (Ø36)
PLA+ · plate other 2
`STL/2026-09-24-lever-knob.stl`

P23 ×1



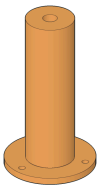
Slew spool (Ø110)
PLA+ · plate other 3
`STL/2026-09-24-slew-spool.stl`

P24 ×1



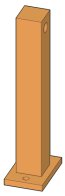
Slew wheel (Ø184)
PLA+ · plate other 2
`STL/2026-09-24-slew-wheel.stl`

P25 ×1



Spool riser
PLA+ · plate other 2
`STL/2026-09-24-spool-riser.stl`

P26 ×2



Slew sleeve post
PLA+ · plate other 2
`STL/2026-09-24-slew-tube-post.stl`

USE THE CURRENT FILES FOR P09, P10 AND P12. The base was rebuilt on 2026-09-29 (no supports needed under the race rim, the four unused outer holes filled) and both bucket ears were rebuilt as clean CAD solids (reinforced joint, no mesh defects). The slicer projects in `3MF/` still contain the older versions: replace them before printing plates 4 and 6 (Stage 1). Already printed? Use what you have. Do not use the superseded patched-mesh files `STL/2026-09-29-bucket-ear-left.stl` / `-right.stl`.

ARM-SHELL HALVES are mirror images. The **left** halves (P01, P03) sit on the +v side, the operator's left when the arm points away from the box.

PARTS IDENTIFICATION

BOARDS, WOOD PIECES AND STOCK

23 wood pieces from the boards you have. Every size is worked out from the minimal CAD model for 7/16″ OSB and the 5/8″ particle board VERIFIED; cut sizes, board layouts and hole positions are all in Stage 2.

YOUR BOARDS

Board	Qty	Use
7/16″ OSB, 2 × 4 ft	3	2 boards cut into 22 pieces; the 3rd is spare
5/8″ particle board, 2 × 2 ft	1	sandbox floor W17, used whole (no cuts)

WOOD PIECES BY GROUP

Group	IDs	Pieces	sq in
Control box	W01, W02, W03, W04, W05, W06, W07, W08	8	509
Pedestal	W09, W10, W11, W12, W13, W14, W15, W16	8	278
Sandbox	W17, W18, W19, W20, W21	5	1416
Spacer frames	W22, W23	2	98

STOCK ITEMS

ID	Item	Qty / length	Used for
S01	1/16″ galvanized 7×19 wire rope	5 pieces	49-13/16″ / 67″ / 80-7/8″ / 30-5/16″ / 33-1/16″
S02	1/16″ single crimp sleeve	5	3 joint-drum midpoints + 2 slew rope ends
S03	1/16″ double crimp sleeve	8	6 arm tail loops + 2 slew loops
S04	PTFE tube 4 mm OD × 2 mm ID	6 pieces	23-1/8″ / 23-1/2″ / 29-5/8″ / 30″ / 10-15/16″ / 13-5/8″
S05	6 mm airsoft BB (or 1/4″ steel ball)	57	slewing ring (+ about 5 spare)
S06	3/4″ PEX-B tube	4-3/4″	turret tube (turns with the turret)
S07	1/2″ copper 90° sweep elbow + 2″ of pipe	1	fixed elbow in the pedestal; deburr both ends
S08	7/16″ OSB, 2 × 4 ft panel	3 (2 used)	all panels except the sandbox floor (Stage 2)
S09	5/8″ particle board, 2 × 2 ft	1	sandbox floor W17, used whole
S10	Wood glue; CA or epoxy; silicone; cable tie	—	panel joints + spacers; ears, knobs, elbow post; spacer seal; elbow

IDS W01–W21 are the same pieces as in the full-design manual; W22 and W23 are the spacer frames that replace the conduit. Pencil the ID on every piece as you cut it.

STAGE 1

PRINT AND CHECK THE PARTS

Your parts are already printed: do each plate's fit test now, before a bad print is buried in the assembly.

CONFIGURATION: SLICER SETTINGS VERIFIED

STORED IN THE PROJECT FILES — OPEN, CHECK, PRINT

Printer preset	**A1 0.6 Fast Start (your preset)**
Process	**0.30mm Strength @BBL A1 0.6 nozzle — 0.3 mm layers, 4 walls**
Infill	**25 % (40 % on drums, hubs, clamps and pinned parts)**
Supports	**tree, only where the plate marks them**
PETG filament	**Generic PETG @BBL A1 — arm shell only**
PLA+ filament	**SUNLU PLA+ 2.0 @BBL A1 — everything else**
PLA+ speed (project)	**flow cap 28 mm³/s at 230 °C; inner walls / infill / supports 200 mm/s; accel 10000**
If infill looks thin	**lower the filament's max volumetric speed to 25**

Fig. 6 Summary of the settings saved in `3MF/2026-09-27-{bambu,orca}-*.3mf` (not a screenshot: your slicer shows them under the printer, process and filament presets).

1. Open the project for your slicer from `3MF/`: `2026-09-27-bambu-…` for Bambu Studio 2.8.2.61 or `2026-09-27-orca-…` for OrcaSlicer. Each opens with its plates named and the settings in place.
2. In the PLA+ project, replace the base on plate 4 with `STL/2026-09-29-base.stl` and both ears on plate 6 with `STL/2026-09-29-bucket-ear-{left,right}-cad.stl`. Keep their orientation: base flat side down; ears on their outer side.
3. Every part is already in its print orientation and every wall was checked for the 0.6 mm nozzle — do not rotate or rescale parts.
4. Print the plates in the order of the table below and do each fit test (last column). Parts you already have: do the fit test now. The minimal build uses every part as printed; only the elbow support (P16) gets its base plate sawn off in Stage 4.

Material	Plate	Name	Parts	Fit test after printing
PETG	arm 1	Arm shell	P02, P01, P04, P03	bucket drum-axle journals turn freely in both stick-half nose holes
PLA+	other 1	Pedestal	P17, P14, P15, P16	copper elbow pushes into P17's socket; a PTFE tube slides through its window; the PEX tube turns in P15; P14 slides onto the PEX tube (clamp open)
PLA+	other 2	Control box 1/2	P24, P20, P19, P25, P22 ×3, P26 ×2	5/16″ rod slides through each hub and into its handle socket; rod enters each knob; riser sits flat
PLA+	other 3	Control box 2/2	P18, P23, P21	rod through the hub; spool turns on the rod; 4 PTFE tubes seat together in P18's window and stop on its floor
PLA+	other 4	Turret + slewing ring 1/2	P12 (new file), P11	PEX tube slides into the tower's centre bore
PLA+	other 5	Turret + slewing ring 2/2	P13	a BB rolls freely in the race between the base rim and the ring
PLA+	other 6	Arm mechanism + bucket	P08, P05, P06, P10 + P09 (new files), P07	each drum sits flat

✓ **CHECK AFTER EVERY PLATE.** Remove all support, then do the fit test. A hub that will not slide on the rod, or a drum that rocks on the table, gets reprinted now — not after it is rigged.

ⓘ **NEW BASE (P12).** Only the eight nut-slot roofs need support (about 2 cm²; the race rim has its own 45° support built in). The bill of materials still lists 28 cm² for the old base. VERIFIED

✕ **EARS SWAPPED.** P09 (left) and P10 (right) are mirror images. Mark them L and R as soon as they come off the plate.

STAGE 2

CUT, DRILL AND LABEL

Boards, rods, rope and tube, all in inches. Label every piece the moment it is cut.

WIRE ROPE, 1/16" 7×19 DERIVED

Piece	Cut	Ends
Boom circuit	**49-13/16"**	single crimp at the midpoint (24-7/8"), loop at each end
Stick circuit	**67"**	single crimp at the midpoint (33-1/2"), loop at each end
Bucket circuit	**80-7/8"**	single crimp at the midpoint (40-7/16"), loop at each end
Slew rope A	**30-5/16"**	single crimp at one end (pedestal drum), loop at the other (spool)
Slew rope B	**33-1/16"**	single crimp at one end (pedestal drum), loop at the other (spool)

Each includes 2-3/8" per crimp loop. Total 261" (21.8 ft). Cut 3/4" long if unsure and trim when you crimp.

PTFE TUBE, 4 × 2 MM DERIVED

Piece	Cut	Runs
stick_hi	**23-1/8"**	box fitting → stop in the boom tip
stick_lo	**23-1/2"**	box fitting → stop in the boom tip
bucket_hi	**29-5/8"**	box fitting → stop in the stick nose
bucket_lo	**30"**	box fitting → stop in the stick nose
slew sleeve A	**10-15/16"**	post A → pedestal fitting
slew sleeve B	**13-5/8"**	post B → pedestal fitting

Each includes 3/8" to trim at fitting. Total 130-13/16" (10.9 ft).

RODS, TUBE AND PIPE VERIFIED

Stock	Cut	For
5/16" rod	**8-1/16"**	lever axle (H14)
5/16" rod	**10-13/16"**	slew spool axle (H15)
5/16" rod	**3 × 5-11/16"**	lever handles (H16)
8-32 threaded rod	**4 × 3-1/16"**	through-rods (box fitting → pedestal fitting) (H06)
8-32 threaded rod	**2 × 1-5/8"**	boom split pins (H10)
8-32 threaded rod	**2 × 7/8"**	stick split pins (H11)
8-32 threaded rod	**2-9/16"**	bucket pin (H12)
3/4" PEX-B tube	**4-3/4"**	turret tube (S06)
1/2" copper pipe	**2"**	on the elbow (if your elbow has no leg) (S07)

5/16" rod total 36" plus 4 saw cuts: a 3 ft rod is about 1/4" short, so buy 4 ft. 8-32 total 19-7/8" from your 3 ft Hillman rod.

1. Cut all rope and PTFE pieces. Wrap tape round the rope at the cut mark and cut through the tape.
2. Write the piece name on a tape flag at **both** ends of every rope and tube (e.g. "STICK — box end").
3. On the three arm ropes, mark the midpoint with a second colour.
4. Cut the rods square; file a small chamfer on each end. Run a nut on and off each 8-32 piece to clean the thread.
5. Cut the PEX and copper square and deburr inside and out: a burr will cut a rope.

ROPE ENDS. Tape every cut end until it is crimped. Loose wires go through skin easily.

OPEN ITEM 4: AXLE NUTS. The 8-1/16" lever axle and the 10-13/16" spool axle carry 5/16" nuts. If your 5/16" rod is smooth, thread about 5/8" at each end of those two rods with a 5/16-18 die, or use 5/16" shaft collars.

STAGE 2 — CONTINUED: CUT LIST

Particle board
sandbox floor W17: the whole 24″ × 24″ panel, no cuts

2 OSB boards
22 pieces, 1725 sq in (75 % of the 2 boards)

1 OSB board spare
keep it for re-cuts

Sizes for 7/16″ OSB, worked out from the minimal CAD panels VERIFIED. Tick each piece as you cut it, pencil its ID on it and draw an arrow to its **bottom** edge (holes are measured from it).

	ID	Piece	Cut size (in)	mm	Holes	sq in
☐	W01	box floor	**15″ × 8-7/16″**	381 × 214	1	126
☐	W02	box front	**15″ × 5-1/4″**	381 × 134	7	79
☐	W03	box rear	**15″ × 5-1/4″**	381 × 134	—	79
☐	W04	box right side	**7-9/16″ × 5-1/4″**	192 × 134	—	40
☐	W05	box left side	**7-9/16″ × 5-1/4″**	192 × 134	—	40
☐	W06	box top (removable)	**15″ × 8-7/16″**	381 × 214	4	126
☐	W07	axle bearing block left	**4″ × 2-3/8″**	101 × 60	1	9
☐	W08	axle bearing block right	**4″ × 2-3/8″**	101 × 60	1	9
☐	W09	pedestal top	**7-3/8″ × 7-1/16″**	188 × 180	5	52
☐	W10	pedestal wall -u (conduit)	**7-1/16″ × 6-7/8″**	180 × 175	5	49
☐	W11	pedestal wall +u (removable access)	**7-1/16″ × 6-7/8″**	180 × 175	—	49
☐	W12	pedestal wall -v	**6-7/8″ × 6-1/2″**	175 × 166	—	45
☐	W13	pedestal wall +v	**6-7/8″ × 6-1/2″**	175 × 166	—	45
☐	W14	pedestal shelf	**6-3/16″ × 5-7/16″**	158 × 138	1	34
☐	W15	pedestal shelf cleat -v	**5-7/16″ × 7/16″**	138 × 11	—	2
☐	W16	pedestal shelf cleat +v	**5-7/16″ × 7/16″**	138 × 11	—	2
☐	W17	sandbox floor	**24″ × 24″**	610 × 610	—	576
☐	W18	sandbox wall -u (box side)	**24″ × 8-7/8″**	610 × 226	7	214
☐	W19	sandbox wall +u	**24″ × 8-7/8″**	610 × 226	—	214
☐	W20	sandbox wall -v	**23-1/8″ × 8-7/8″**	588 × 226	—	206
☐	W21	sandbox wall +v	**23-1/8″ × 8-7/8″**	588 × 226	—	206
☐	W22	spacer frame 1	**7-1/16″ × 6-7/8″**	180 × 175	1	49
☐	W23	spacer frame 2	**7-1/16″ × 6-7/8″**	180 × 175	1	49

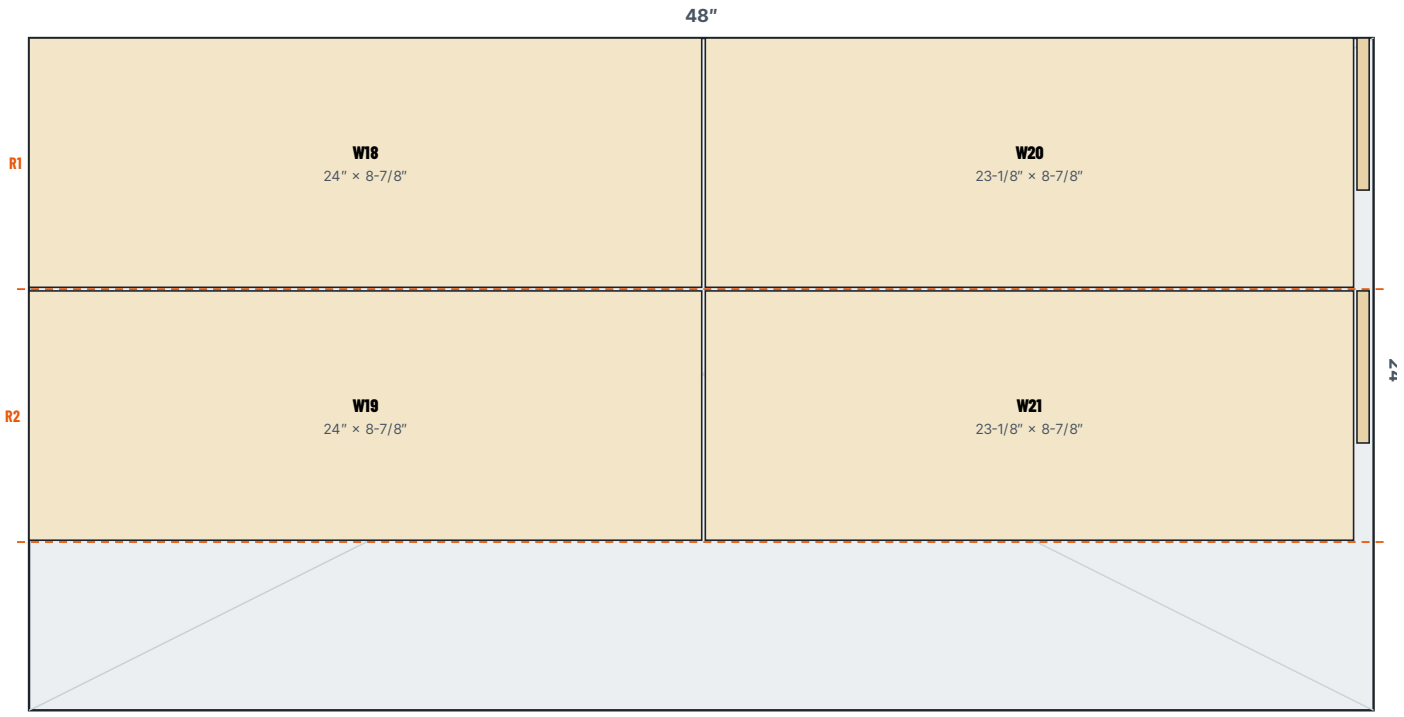
Walls of the sandbox and the pedestal are 5/32″ shorter than the full design: the 5/8″ floor stands on the table with no shim, so the turret and both fittings stay at their design heights. DERIVED

STAGE 2 — CONTINUED: BOARD LAYOUTS

Rip the strips first (orange dashed lines, R1, R2 …), then crosscut each strip. Drawn to scale with a 1/8″ kerf; grey is offcut. After the crosscut, trim any piece that is narrower than its strip to its cut-list size.

CLEATS W15 AND W16. A 7/16″ × 7/16″ stick is too thin to rip safely on its own. Rip it off the edge of a wider offcut with a push stick, or use any square stick at least 7/16″ thick. ASSUMPTION

OSB BOARD 1 — 2 × 4 FT, 7/16″



Thin strips not labelled on the drawing: W15, W16; see the cut order.

R1: rip a strip **8-7/8″** wide; crosscut: 24″ → **W18**; 23-1/8″ → **W20**; 7/16″ → **W15**.

R2: rip a strip **8-7/8″** wide; crosscut: 24″ → **W19**; 23-1/8″ → **W21**; 7/16″ → **W16**.

STAGE 2 — CONTINUED: BOARD LAYOUTS (CONTINUED)

OSB BOARD 2 — 2 × 4 FT, 7/16"

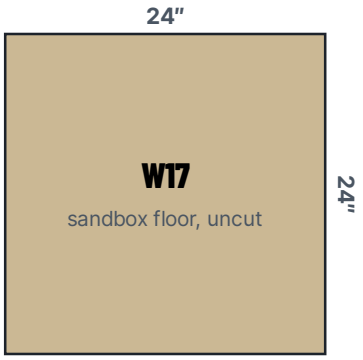


R1: rip a strip **8-7/16"** wide; crosscut: 15" → **W01**; 15" → **W06**; 7-3/8" → **W09**; 7-1/16" → **W10**; 2-3/8" → **W07** + **W08** (rip that piece into 4" / 4").

R2: rip a strip **6-7/8"** wide; crosscut: 7-1/16" → **W11**; 7-1/16" → **W22**; 7-1/16" → **W23**; 6-7/8" → **W12**; 6-7/8" → **W13**; 6-3/16" → **W14**.

R3: rip a strip **5-1/4"** wide; crosscut: 15" → **W02**; 15" → **W03**; 7-9/16" → **W04**; 7-9/16" → **W05**.

PARTICLE BOARD — 2 × 2 FT, 5/8" · AND OSB BOARD 3



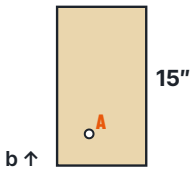
W17 sandbox floor: use the whole particle-board panel, no cuts. The design is 1/64" bigger; that does not matter. Seal both faces and all edges before you build on it (Stage 3).

OSB board 3: not needed. Keep it for a re-cut.

STAGE 2 — CONTINUED: DRILLING

Measure **a** along the bottom edge and **b** up the side, from the lower-left corner as drawn. The bold size is the drill bit. Clamp a scrap under OSB so the holes do not tear out.

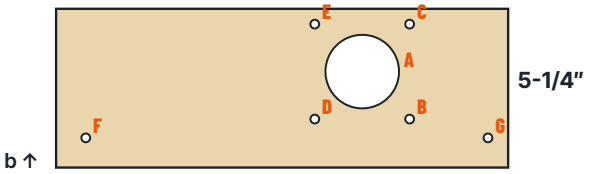
W01 box floor



a from the −u edge, b from the −v (right) edge — seen from above

	Hole / bit	a / b (in)	mm	For
A	**11/32″** Ø8.5 mm	**3-1/16″ / 3″**	77 / 76	spool axle; 7/8″ counterbore 9/32″ deep from below

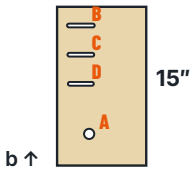
W02 box front



a from the −v (right) edge, b from the bottom edge — seen from the +u side

	Hole / bit	a / b (in)	mm	For
A	**2-7/16″ hole saw** Ø62.0 mm	**10-3/16″ / 3-3/16″**	258 / 81	tube passage
B	**3/16″** Ø4.6 mm	**11-3/4″ / 1-5/8″**	298 / 41	8-32 through-rod
C	**3/16″** Ø4.6 mm	**11-3/4″ / 4-3/4″**	298 / 121	8-32 through-rod
D	**3/16″** Ø4.6 mm	**8-9/16″ / 1-5/8″**	218 / 41	8-32 through-rod
E	**3/16″** Ø4.6 mm	**8-9/16″ / 4-3/4″**	218 / 121	8-32 through-rod
F	**3/16″** Ø4.6 mm	**1″ / 1″**	25 / 25	M3 × 30 corner bolt
G	**3/16″** Ø4.6 mm	**14-5/16″ / 1″**	364 / 25	M3 × 30 corner bolt

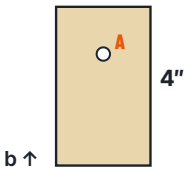
W06 box top (removable)



a from the −u edge, b from the −v (right) edge — seen from above

	Hole / bit	a / b (in)	mm	For
A	**1″ spade / Forstner** Ø24.4 mm	**3-1/16″ / 3″**	77 / 76	spool tube bearing
B	**slot 2-5/8″**: 23/64″ holes at a 1-1/8″ & 3-3/8″, saw between 66.1 mm	**2-1/4″ / 13-1/4″**	57 / 336	boom lever slot = hard stop (42 deg swing)
C	**slot 2-1/2″**: 23/64″ holes at a 1-3/16″ & 3-5/16″, saw between 64.1 mm	**2-1/4″ / 10-1/2″**	57 / 266	stick lever slot = hard stop (41 deg swing)
D	**slot 2-7/16″**: 23/64″ holes at a 1-3/16″ & 3-5/16″, saw between 61.8 mm	**2-1/4″ / 7-11/16″**	57 / 196	bucket lever slot = hard stop (40 deg swing)

W07 + W08 axle bearing blocks (×2, identical)



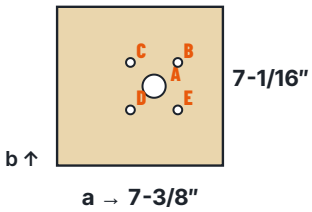
a from the −u edge, b from the bottom edge — seen from the −v side

	Hole / bit	a / b (in)	mm	For
A	**11/32″** Ø8.5 mm	**1-3/16″ / 2-13/16″**	30 / 71	5/16″ lever axle

! **DRILL THE BOX FRONT W02 THROUGH THE SANDBOX WALL W18.** Their four rod holes and two corner holes must line up, but the panels are not the same size. Drill W18 from its card. Clamp W02 on it face to face, W02's −v end 1-9/16″ in from W18's −v end and W02's bottom edge 3/16″ below W18's (the two 2-7/16″ holes then line up), and drill the six 3/16″ holes through W18's holes into W02. DERIVED

STAGE 2 — CONTINUED: DRILLING (CONTINUED)

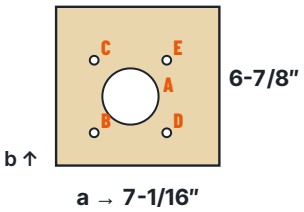
W09 pedestal top



a from the −u edge, b from the −v (right) edge — seen from above

	Hole / bit	a / b (in)	mm	For
A	**1″ spade / Forstner** Ø26.2 mm	**4-5/16″ / 3-9/16″**	110 / 90	PEX turret tube clearance
B	**1/8″** Ø3.0 mm	**5-3/8″ / 4-5/8″**	137 / 117	pilot, #8 screw from the base
C	**1/8″** Ø3.0 mm	**3-1/4″ / 4-5/8″**	83 / 117	pilot, #8 screw from the base
D	**1/8″** Ø3.0 mm	**3-1/4″ / 2-1/2″**	83 / 63	pilot, #8 screw from the base
E	**1/8″** Ø3.0 mm	**5-3/8″ / 2-1/2″**	137 / 63	pilot, #8 screw from the base

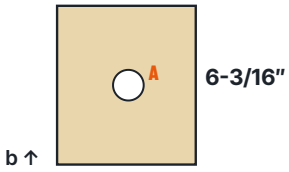
W10 pedestal wall -u (conduit)



a from the −v (right) edge, b from the bottom edge — seen from the +u side

	Hole / bit	a / b (in)	mm	For
A	**2-7/16″ hole saw** Ø62.0 mm	**3-1/4″ / 3″**	82 / 76	tube passage
B	**3/16″** Ø4.6 mm	**1-5/8″ / 1-7/16″**	42 / 36	8-32 through-rod
C	**3/16″** Ø4.6 mm	**1-5/8″ / 4-9/16″**	42 / 116	8-32 through-rod
D	**3/16″** Ø4.6 mm	**4-13/16″ / 1-7/16″**	122 / 36	8-32 through-rod
E	**3/16″** Ø4.6 mm	**4-13/16″ / 4-9/16″**	122 / 116	8-32 through-rod

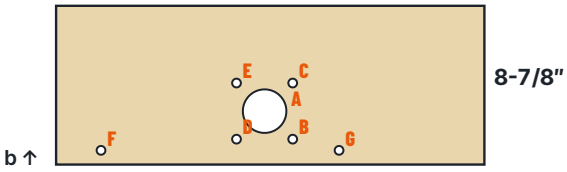
W14 pedestal shelf



a from the −u edge, b from the −v (right) edge — seen from above

	Hole / bit	a / b (in)	mm	For
A	**1-1/4″ spade / Forstner** Ø30.4 mm	**2-13/16″ / 3-1/8″**	71 / 79	PEX bushing P15

W18 sandbox wall -u (box side)

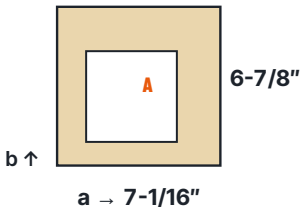


a from the −v (right) edge, b from the bottom edge — seen from the +u side

	Hole / bit	a / b (in)	mm	For
A	**2-7/16″ hole saw** Ø62.0 mm	**11-11/16″ / 3″**	297 / 76	tube passage
B	**3/16″** Ø4.6 mm	**13-1/4″ / 1-7/16″**	337 / 36	8-32 through-rod
C	**3/16″** Ø4.6 mm	**13-1/4″ / 4-9/16″**	337 / 116	8-32 through-rod
D	**3/16″** Ø4.6 mm	**10-1/8″ / 1-7/16″**	257 / 36	8-32 through-rod
E	**3/16″** Ø4.6 mm	**10-1/8″ / 4-9/16″**	257 / 116	8-32 through-rod
F	**3/16″** Ø4.6 mm	**2-1/2″ / 13/16″**	64 / 20	M3 × 30 corner bolt
G	**3/16″** Ø4.6 mm	**15-7/8″ / 13/16″**	403 / 20	M3 × 30 corner bolt

STAGE 2 — CONTINUED: DRILLING (CONTINUED)

W22 + W23 spacer frames (×2, identical)



a from the −v (right) edge, b from the bottom edge — seen from the +u side

	Hole / bit	a / b (in)	mm	For
A	**window 3-15/16″ square**: corners at a 1-1/4″ – 5-3/16″, b 1″ – 4-15/16″ 100 mm	**3-1/4″ / 3″**	82 / 76	window (3/8″ hole in each corner, then jigsaw)

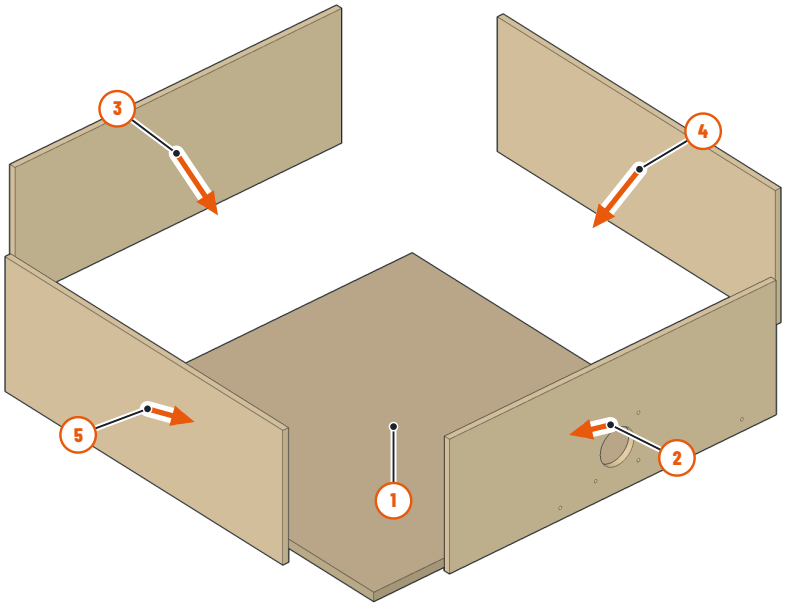
Drilling tips ASSUMPTION

- **Clean holes in OSB:** clamp a scrap under the panel, or drill until the point shows and finish from the other side.
- **2-7/16″ holes:** hole saw with its pilot; cut halfway from each face.
- **Lever slots (W06):** drill a 23/64″ hole at each end position given, saw or jigsaw the two sides between them, file smooth. The handle must run end to end without catching: the slot ends are the lever stops.
- **Spacer windows (W22, W23):** mark the square, drill a 3/8″ hole inside each corner, jigsaw between them. Slightly big is fine; never smaller.
- **Counterbore under the box floor (W01):** 7/8″ Forstner 9/32″ deep first, then the 11/32″ hole through its centre.

STAGE 3

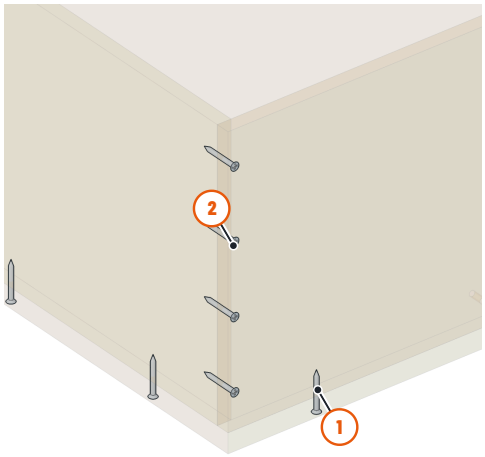
BUILD THE SANDBOX

Particle-board floor and four OSB walls, glued and screwed. The −u wall is the drilled one: the control box bolts to its outside.



1. Sandbox floor: 24″ × 24″ particle board
2. Wall −u (box side, drilled)
3. Wall +u
4. Wall −v
5. Wall +v

Fig. 7 Sandbox: four walls on the floor.



1. #6 × 1-1/4″ up through the floor
2. #6 × 1-1/4″ through wall −u into wall +v

Fig. 8 Corner joint, walls see-through.

YOU NEED

W17	Floor, 5/8″ particle board 24″ × 24″	1
W18	Wall −u (drilled), 24″ × 8-7/8″	1
W19	Wall +u, 24″ × 8-7/8″	1
W20	Wall −v, 23-1/8″ × 8-7/8″	1
W21	Wall +v, 23-1/8″ × 8-7/8″	1
H20	#6 × 1-1/4″ wood screw	32
	Wood glue; floor sealer	

1. Seal the particle board on both faces and all four edges and let it dry. ASSUMPTION
2. Glue and screw the four walls **on top of** the floor: 4 screws up through the floor into each wall's bottom edge. Drill a 3/32″ pilot for every screw into an OSB edge, and keep screws 1/2″ from the corners.
3. The −v and +v walls fit between the −u and +u walls: 4 screws through each ±u wall into the ends of the ±v walls.
4. Put the −u wall's holes at the bottom: the 2-7/16″ hole centre is 3″ above the wall's bottom edge and 11-11/16″ from its −v end (operator's right). VERIFIED
5. Check the box is square (equal diagonals) before the glue sets.

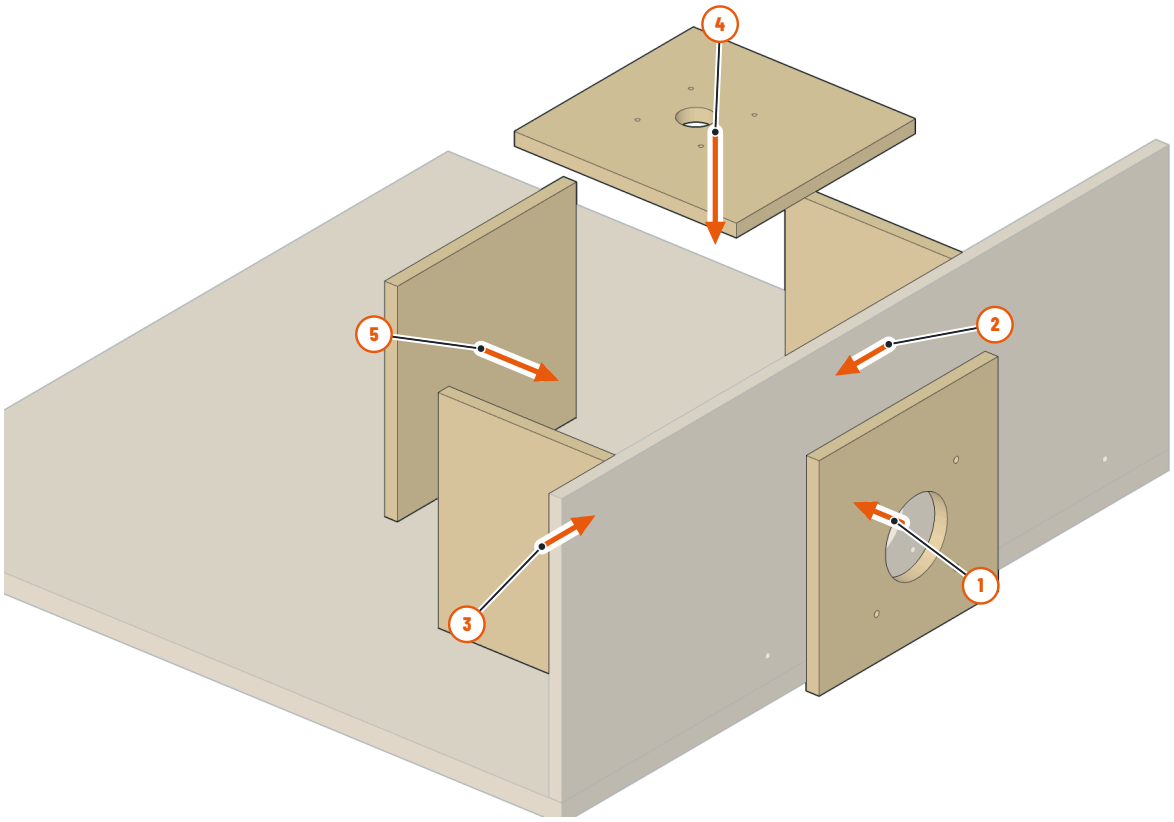
✓ **CHECK.** Walls flush with the floor edges; diagonals equal within 1/16″; walls 8-7/8″ tall, so the rim is 9-1/2″ above the table.

✕ **HOLE SIDE.** Fitting the −u wall upside down puts the big hole near the top. The hole is 3″ from the **bottom**.

STAGE 4

BUILD THE PEDESTAL

An OSB column that stands against the sandbox wall on two spacer frames. It carries the slewing ring on top and the slew drum inside.



1. Wall −u (conduit wall)
2. Wall −v
3. Wall +v
4. Pedestal top
5. Wall +u (removable — screw only)

Fig. 9 Pedestal walls and top. Wall +u is screwed only: it is the service hatch.

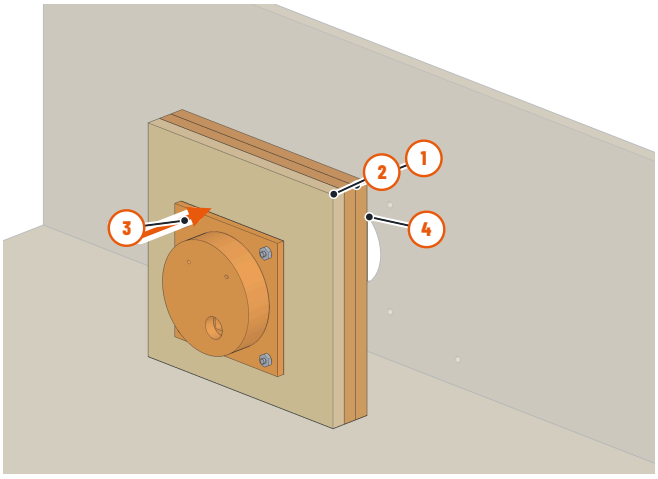
YOU NEED

W09	Pedestal top, 7-3/8″ × 7-1/16″	1
W10	Wall −u (drilled), 7-1/16″ × 6-7/8″	1
W11	Wall +u (removable), 7-1/16″ × 6-7/8″	1
W12	Wall −v, 6-7/8″ × 6-1/2″	1
W13	Wall +v, 6-7/8″ × 6-1/2″	1
H20	#6 × 1-1/4″ wood screw	32

1. Glue and screw walls −v and +v between walls −u and +u (4 screws per joint, through the ±u walls, 3/32″ pilots). Fit wall +u **with screws only, no glue**.
2. Glue and screw the top onto the four walls (4 screws per wall). The top's 1″ hole is the PEX clearance; its four 1/8″ holes are pilots for the base screws.
3. Build the pedestal on the bench, not in the sandbox: the spacers and the rods go on first (next page). The figure shows it in its final place.

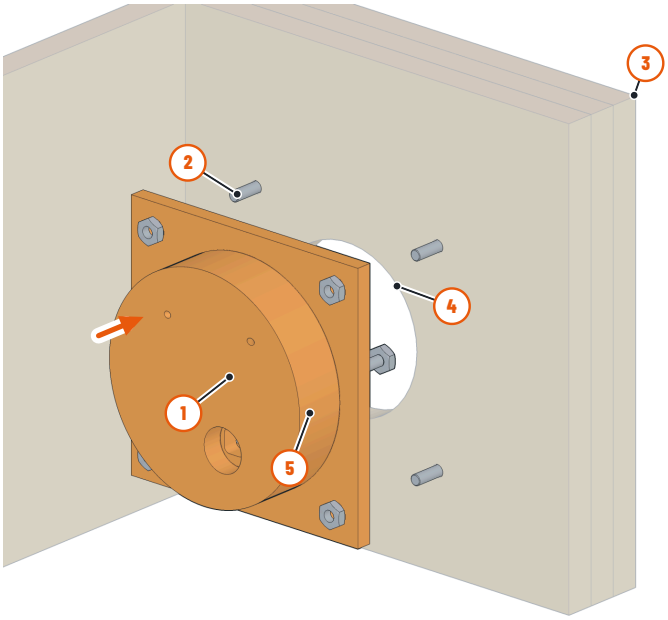
✓ **CHECK.** The pedestal is 7-5/16″ tall and its top is flat and square to the walls. Wall −u has a 2-7/16″ hole and four 3/16″ holes around it.

STAGE 4 — CONTINUED: SPACER FRAMES, PEDESTAL FITTING AND THROUGH-RODS



1. Spacer frames W22 + W23, glued to the pedestal wall
2. Pedestal wall –u (W10)
3. 8-32 rods, nuts on the pedestal fitting
4. Rods go through the four 3/16″ holes in the sandbox wall

Fig. 10 The stack that replaces the conduit (cut through the middle).



1. Pedestal conduit fitting (P17)
2. 4 × 8-32 rod 3-1/16″, nut on the fitting
3. Spacer frames W22 + W23 (glued on)
4. Wall –u
5. Copper socket (elbow)

Fig. 11 P17 on wall –u with the four rods, from inside the pedestal.

YOU NEED (THIS PAGE)

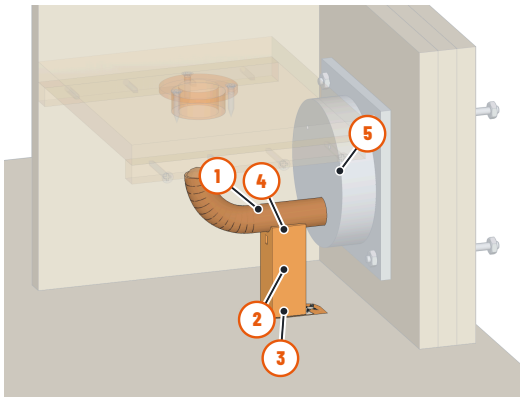
W22	Spacer frame 1	1
W23	Spacer frame 2	1
	Wood glue	
P17	Pedestal conduit fitting	1
H06	8-32 rod, 3-1/16″	4
H13	8-32 nut	4

WHY SPACERS. Two 7/16″ frames fill the gap the conduit used to cross. Their 3-15/16″ windows, with the holes in the walls, make a sealed 3-1/8″ chamber between the two fittings. DERIVED

4. Glue the two spacer frames to the **outside** of wall –u, one on the other, edges flush with the wall's edges, windows lined up. The four 3/16″ holes and the 2-7/16″ hole must all be clear inside the window. Clamp until set.
5. From inside the pedestal, hold P17's square flange flat against wall –u: the big socket faces the wall's hole; the round boss, with the small copper socket in its end, points into the pedestal. Line up its four corner holes with the wall's.
6. Push a 3-1/16″ 8-32 rod through each corner: P17's flange, the wall, then out through the spacer window. Run an 8-32 nut onto the inner end until about 1/16″ of thread shows past it, and snug it against the flange. Each rod now sticks out 1-5/16″ past the spacers.
7. Keep the four 8-32 nuts for the box side in this stage's bag (Stage 7).

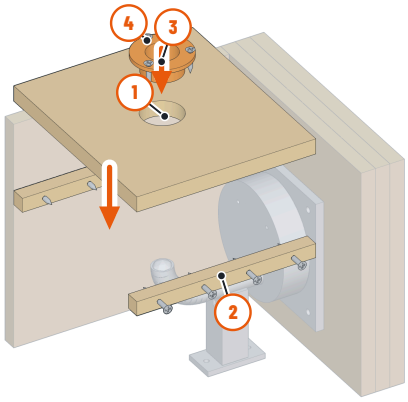
FITTING TURNED ROUND. If the big socket faces into the pedestal, the copper socket faces the wall and the tubes cannot reach the turret.

STAGE 4 — CONTINUED: INTO THE SANDBOX, COPPER ELBOW, SHELF



1. 1/2″ copper elbow (pushed into the fitting socket)
2. Elbow support (P16)
3. Base plate sawn off; 1 × #6 × 1-1/4″ up through the floor
4. Cable-tie slot
5. Pedestal fitting

Fig. 12 Copper elbow on the sawn elbow support.



1. Shelf (rests on the cleats)
2. Shelf cleats (screwed to walls ±v)
3. PEX bushing (P15), flange up
4. 3 × #6 × 3/4″

Fig. 13 Cleats, shelf and PEX bushing (shelf and bushing lifted out).

YOU NEED (THIS PAGE)

S07	1/2″ copper elbow + pipe	1
P16	Elbow support (base plate sawn off)	1
H20	#6 × 1-1/4″ (elbow support)	1
	Silicone; CA or epoxy; cable tie	
W14	Shelf	1
W15	Shelf cleat −v	1
W16	Shelf cleat +v	1
H20	#6 × 1-1/4″ (cleats)	8
P15	PEX bushing	1
H19	#6 × 3/4″ (bushing)	3

7. Run a bead of silicone round the outer face of the outer spacer, just outside its window. Stand the pedestal on the sandbox floor and slide it against the −u wall so the four rods pass through the wall's 3/16″ holes. The rods stick out 7/8″ outside the sandbox. Fit a nut finger-tight on each to hold the pedestal while the silicone cures (they come off again in Stage 7).
8. Saw the 5/32″ base plate off the elbow support, flush with its post. Push the copper elbow's horizontal leg into P17's copper socket; the other leg points straight up.
9. Stand the post under the horizontal leg (about 1-15/16″ in from the inside of wall −u), mark its centre on the floor, drill a 1/8″ pilot through the floor there, and fix it with one #6 × 1-1/4″ up from below plus a few drops of CA. Cable-tie the elbow into the saddle.
10. Screw the cleats to the inside of walls −v and +v, their tops 4-7/16″ above the sandbox floor. Lay the shelf on them and screw the PEX bushing into the shelf hole, flange on top.

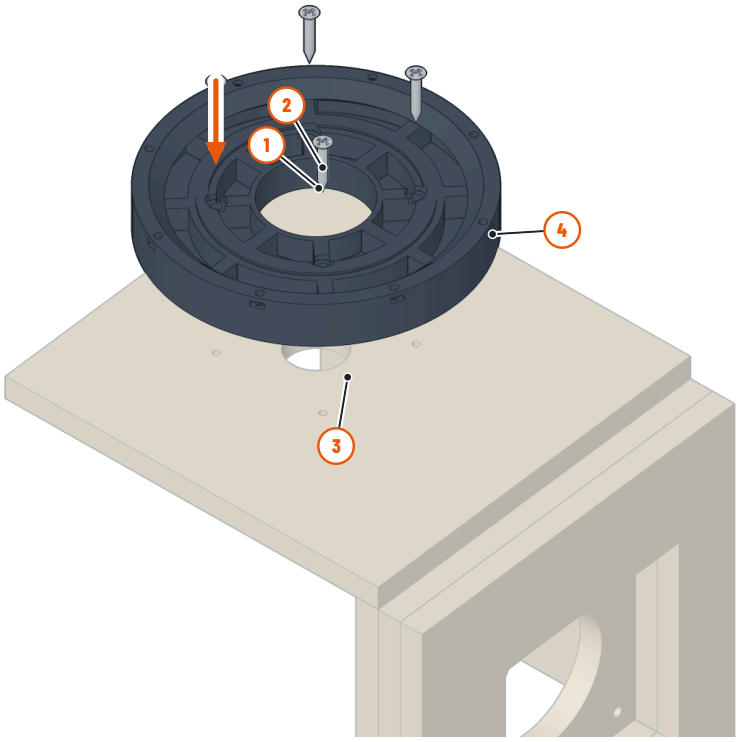
✓ **CHECK.** The elbow's top opening is directly under the bushing's bore (sight down through it). The copper does not move when pulled. No light shows between the outer spacer and the sandbox wall. VERIFIED

? **OPEN ITEM 5: FIXING THE PEDESTAL DOWN.** The rods clamp the pedestal to the sandbox wall once the box is on (Stage 7). For a firmer base, add 4 × #6 × 1-1/4″ up through the floor into the ±v walls, 2 each (not in the bill of materials). ASSUMPTION

STAGE 5

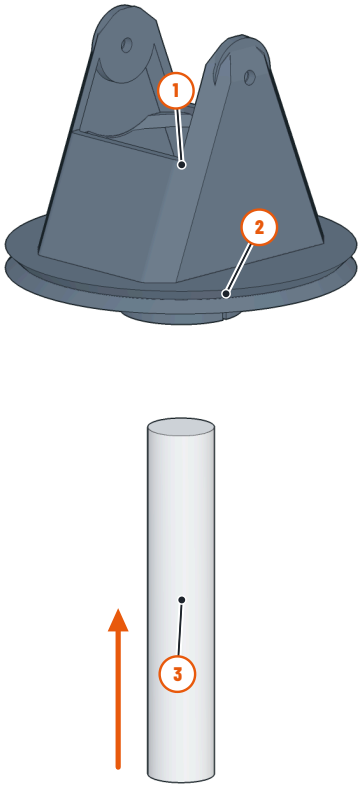
SLEWING RING AND TURRET

The base screws to the pedestal; the turret rides on 57 BBs and is held down by the retaining ring.



1. Base (P12) — current file 2026-09-29-base
2. 4 × #8 × 1″ wood screw in the counterbores
3. Pedestal top (1/8″ pilots)
4. Nut slots in the rim (8)

Fig. 14 Base onto the pedestal top.



1. Tower (P11)
2. Turret hub clamp: M3 × 25 + lock washer + nut
3. 3/4″ PEX-B tube, 4-3/4″

Fig. 15 PEX tube into the turret hub.

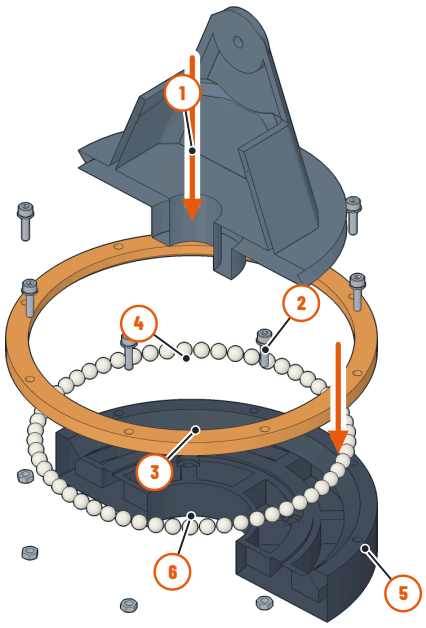
YOU NEED

P12	Base (2026-09-29 file)	1
H18	#8 × 1″ wood screw	4
P11	Tower	1
S06	3/4″ PEX-B tube, 4-3/4″	1
H04	M3 × 25 (hub clamp)	1
H09	M3 split-lock washer	1
H07	M3 nut	1

1. Set the base on the pedestal top with its centre bore over the 1″ hole. Turn it until its four counterbored holes sit over the four 1/8″ pilots.
2. Screw it down with 4 × #8 × 1″ in the counterbores. Do not over-tighten into the OSB.
3. Push the PEX tube up into the tower's clamp hub from below until its top is **3 mm below the top face of the turret flange** (tube top at z 22, flange top at z 25). VERIFIED
4. Tighten the hub's pinch clamp: M3 × 25 through the hub, split-lock washer and nut on the far side. The tube must not turn in the hub.

! **CLAMP THE TUBE BEFORE THE TURRET GOES ON.** Once the turret sits on the balls, the hub clamp is inside the base and cannot be reached. VERIFIED

STAGE 5 — CONTINUED: BALLS, RETAINING RING, SLEW DRUM

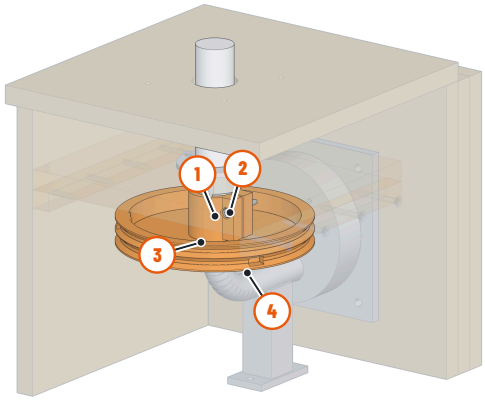


1. Turret (tower + PEX tube) lowered last
2. 8 × M3 × 12 + split-lock washer
3. Retaining ring (P13)
4. 57 × 6 mm BBs in the race
5. 8 × M3 nut, slid into the rim slots
6. Base (P12)

Fig. 16 Slewing ring, exploded and cut in half to show the race.

YOU NEED (THIS PAGE)

S05	6 mm BBs	57
P13	Retaining ring	1
H02	M3 × 12 socket cap	8
H09	M3 split-lock washer	9
H07	M3 nut	9
P14	Pedestal slew drum	1
H04	M3 × 25 (drum clamp)	1



1. Slew drum (P14) on the PEX tube
2. Pinch clamp: M3 × 25 + lock washer + nut
3. PEX tube end 1/16″ below the drum
4. Copper elbow end inside the PEX bore

Fig. 17 Slew drum on the PEX tube inside the pedestal (wall +u removed).

5. Slide an M3 nut into each of the 8 slots in the outside of the base rim, flat side up, until it lines up with the hole above it.
6. Lay the 57 BBs in the base's race (the 45° groove round the rim). They fill the circle with about 0.4 mm between balls.
7. From inside the pedestal (wall +u off; move it aside), hold the slew drum under the shelf, hub up. Lower the turret: the PEX tube goes down through the base, the pedestal top, the bushing and into the drum. The turret flange settles on the balls.
8. Lay the retaining ring on the rim and fit the 8 × M3 × 12, a split-lock washer under each head, into the nuts. Tighten them in a cross pattern, a little at a time.
9. Slide the drum up or down the tube until the tube's end is 2 mm below the drum's underside, and tighten its pinch clamp (M3 × 25, lock washer, nut) lightly. Its rotation is set in Stage 10.

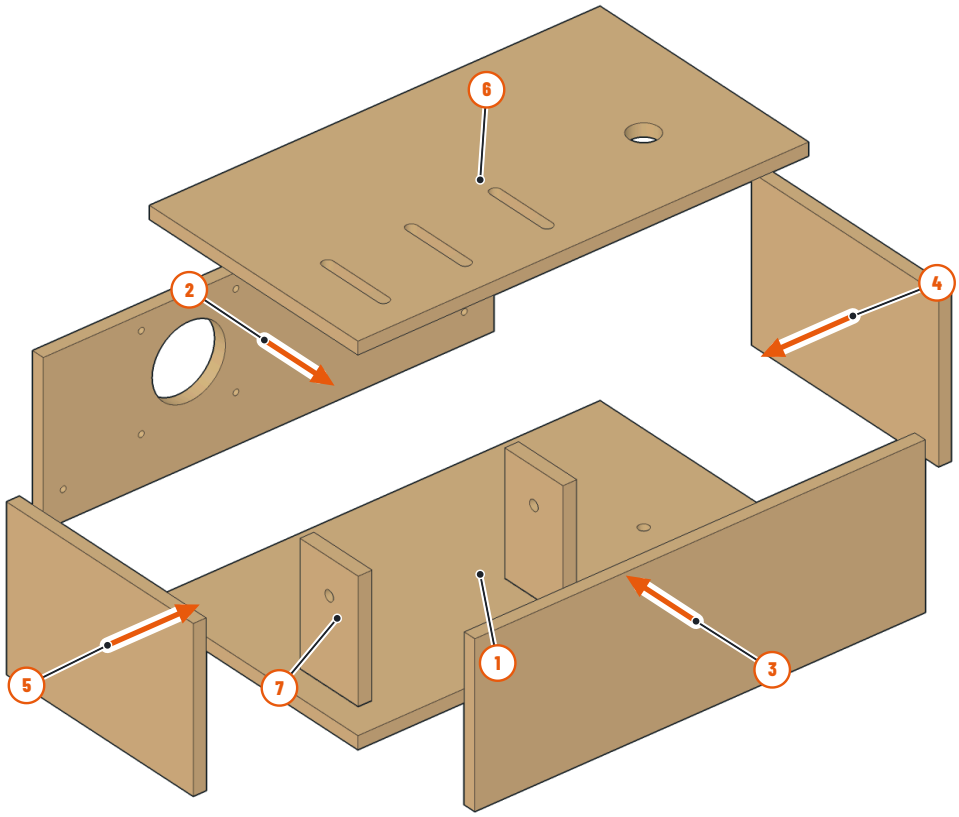
✓ **CHECK — TURRET.** The turret turns by hand with no rocking and no lift. The model allows 0.65 mm lift and 1.1° rock before parts touch. If it rocks, shim under the retaining ring; if it binds, loosen the ring screws evenly a quarter turn. VERIFIED

! **COUNT THE BALLS.** 57 balls. One missing makes a gap that knocks as the turret turns; one extra jams the ring.

STAGE 6

BUILD THE CONTROL BOX

Box carcass, lever axle with its three hubs, the slew spool and the two sleeve posts. The left side and the top go on later.



1. Floor
2. Front (to the sandbox; drilled)
3. Rear
4. Right side
5. Left side (fit after the lever axle)
6. Top (removable, fitted last)
7. Bearing blocks

Fig. 18 Control box panels.

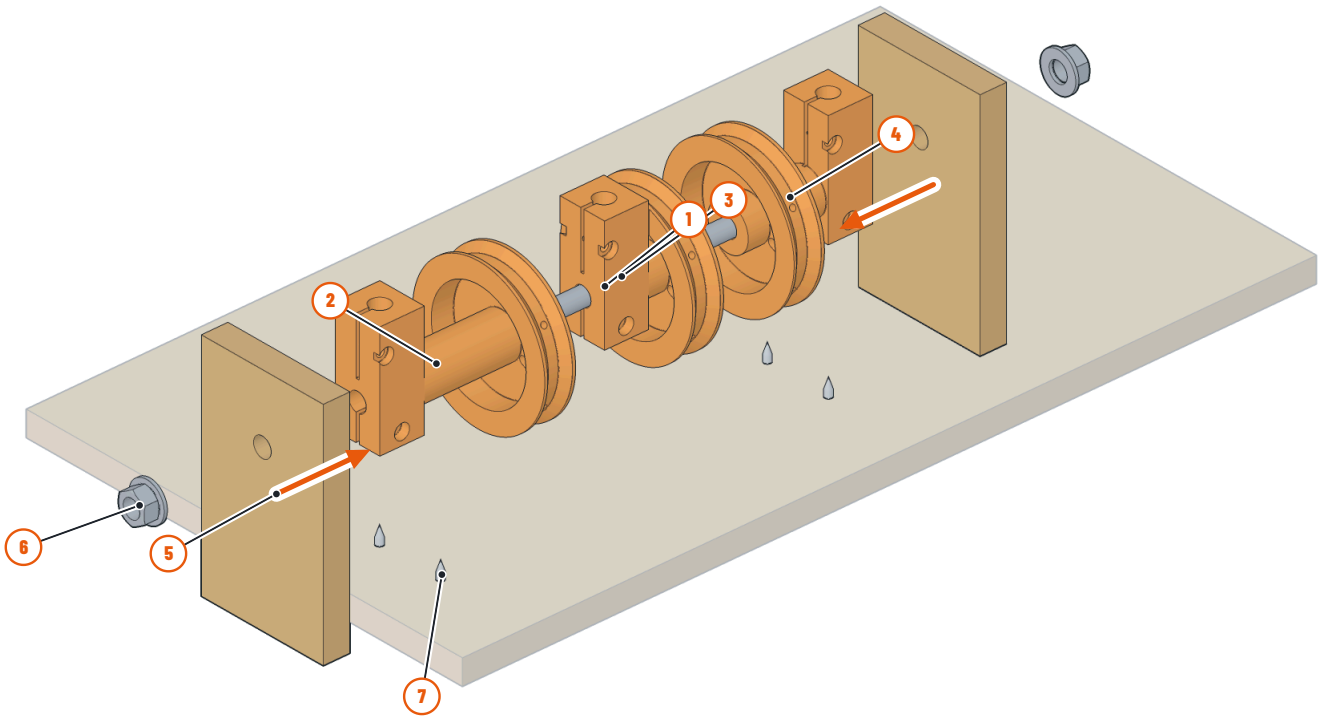
YOU NEED

W01	Box floor, 15″ × 8-7/16″	1
W02	Box front (drilled), 15″ × 5-1/4″	1
W03	Box rear, 15″ × 5-1/4″	1
W04	Box right side, 7-9/16″ × 5-1/4″	1
W05	Box left side (later in this stage)	1
H20	#6 × 1-1/4″ wood screw	32
	Wood glue	

1. Glue and screw the front, rear and right side onto the floor (4 screws per edge, up through the floor, 3/32″ pilots).
2. Join the front and rear to the right side (4 screws per joint, through the front and rear).
3. The front's 2-7/16″ hole is low (centre 3-3/16″ above the floor's top face) and right of centre: 10-3/16″ from the box's right-hand (–v) outside face, 4-13/16″ from its left. VERIFIED
4. Leave the left side and the top off for now.

! **BOX SIZE.** With the hubs and spool you already printed, the box is 8-7/16″ deep, 15″ wide and 6-1/8″ high outside. VERIFIED

STAGE 6 — CONTINUED: LEVER AXLE AND HUBS



1. 5/16″ lever axle, 8-1/16″
2. Boom hub (P19) — operator's left
3. Stick hub (P20) — centre
4. Bucket hub (P21) — right
5. Bearing block (W07/W08)
6. 5/16″ SAE washer + nut, each end
7. 2 × #6 × 3/4″ from below, each block

Fig. 19 Lever axle stack, exploded along the axle: blocks, three hubs, washers and nuts.

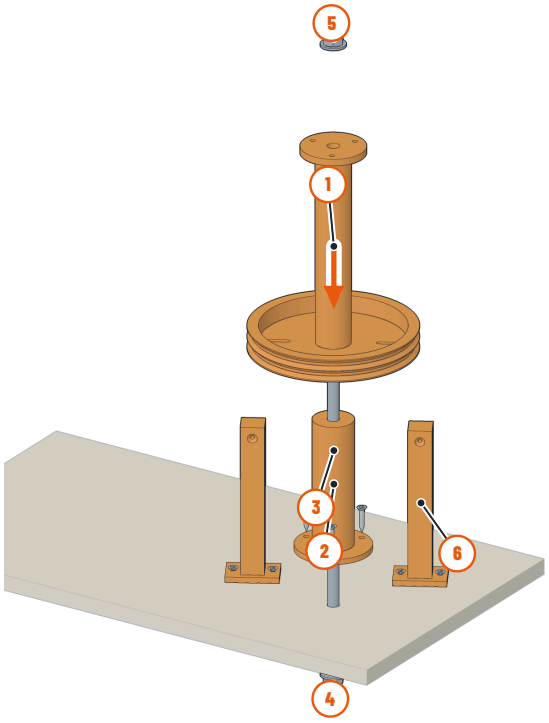
YOU NEED (THIS PAGE)

W07	Axle bearing block left	1
W08	Axle bearing block right	1
H14	5/16″ rod, 8-1/16″	1
P19	Lever hub, boom	1
P20	Lever hub, stick	1
P21	Lever hub, bucket	1
H17	5/16″ nut + SAE washer	2
H19	#6 × 3/4″ wood screw	4

1. On the bench, slide onto the rod in this order from the left (+v) end: left block, **boom** hub, **stick** hub, **bucket** hub, right block. The handle sockets all point the same way (up).
2. Centre the rod (4-1/16″ each side of the middle). Fit a washer and a nut outside each block, finger-tight plus a quarter turn: the hubs must still turn freely. They are 0.5 mm apart by design. VERIFIED
3. Stand the stack on the box floor: axle 2-1/4″ from the floor's rear (−u) edge, left block's outer face 15/16″ in from the floor's left (+v) edge. Screw each block from below: 2 × #6 × 3/4″ (1/8″ pilots in the block edge). VERIFIED
4. Now fit the left side (glue + 4 screws per edge).

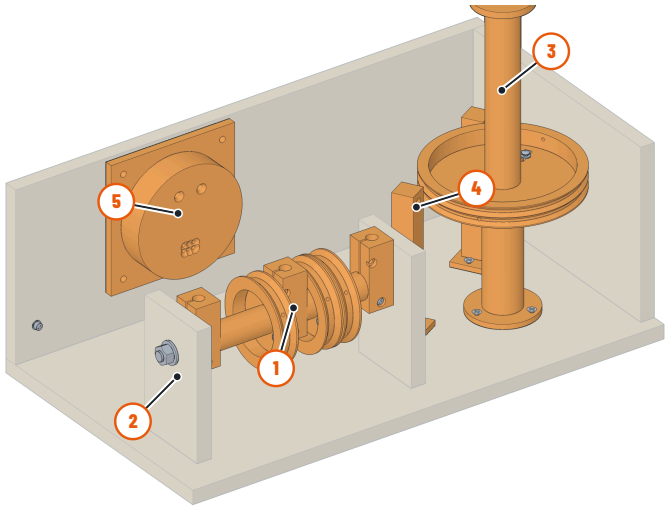
HUB ORDER. Boom (P19) on the operator's left, stick (P20) centre, bucket (P21) right. The hubs look alike — check the label on each before sliding it on. VERIFIED

STAGE 6 — CONTINUED: SLEW SPOOL, RISER AND SLEEVE POSTS



1. Slew spool (P23)
2. Spool riser (P25) + 3 × #6 × 3/4″
3. 5/16″ spool axle, 10-13/16″
4. Washer + jam nut in the 7/8″ counterbore under the floor
5. Washer + nut on top (fitted with the wheel)
6. Sleeve posts (P26 ×2) + 2 × #6 × 3/4″ each

Fig. 20 Spool axle, riser, spool and posts.



1. Lever axle + 3 hubs
2. Bearing blocks
3. Spool riser + slew spool
4. Sleeve posts
5. Box conduit fitting (fitted in Stage 7)

Fig. 21 Inside the box at the end of Stage 6 (top, rear and left side hidden).

YOU NEED (THIS PAGE)

P25	Spool riser	1
P23	Slew spool	1
H15	5/16″ rod, 10-13/16″	1
H17	5/16″ nut + SAE washer	1
P26	Slew sleeve post	2
H19	#6 × 3/4″ wood screw	7
P18	Box conduit fitting (loose for now)	1

POST POSITIONS come from the model. Centres, from the floor's rear (–u) edge and right (–v) edge: post A 3-7/16″ / 5-3/4″, post B 5-5/8″ / 1-15/16″. Each post's counterbore lines up with the spool's drum tangent. VERIFIED

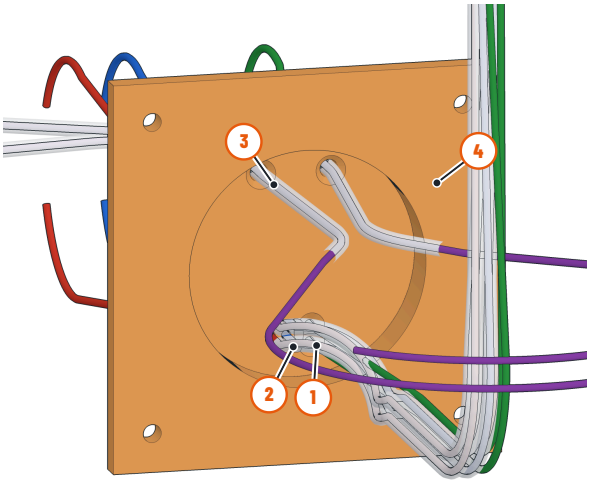
5. Push the 10-13/16″ rod up through the floor's 11/32″ hole from below. Fit a washer and a nut on its bottom end inside the 7/8″ × 9/32″ counterbore under the floor (the model uses a thin jam nut there so it stays flush).
6. Slide the riser down the rod onto the floor and screw it through its flange (3 × #6 × 3/4″).
7. Slide the spool onto the rod, drum end down, onto the riser. There is no washer between them: the gap is 0.5 mm. VERIFIED
8. Screw the two sleeve posts to the floor beside the spool, each with 2 × #6 × 3/4″ through its base. Post **A** is the one nearer the levers. Each post's counterbore faces away from the spool.
9. Put the box conduit fitting (P18) aside — it goes in at Stage 7, together with the PTFE tubes and the rods.

CHECK. Each hub turns freely on the axle and the spool turns freely on its rod. Nothing in the box rubs the hubs.

STAGE 7

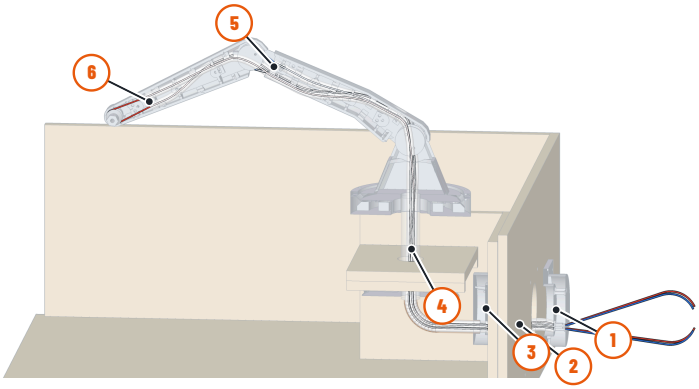
PTFE TUBES AND BOX MOUNTING

The four arm tubes are seated in the box fitting and threaded through the spacer chamber to the turret; then the box goes onto the rods.



1. 4 PTFE tubes seat together in the rounded window
2. Boom rope holes (bare rope)
3. Slew sleeve holes (Ø4.6, pass-through)
4. Conduit socket

Fig. 22 Box fitting P18 from the sandbox side, with the tube and rope paths.



1. Box fitting window (tube stop)
2. Spacer chamber: spare tube length bows here
3. Pedestal fitting window (tubes slide)
4. Up the copper elbow and the PEX tube
5. Stick tubes stop in the boom tip
6. Bucket tubes stop in the stick nose

Fig. 23 Where the four arm tubes run when finished.

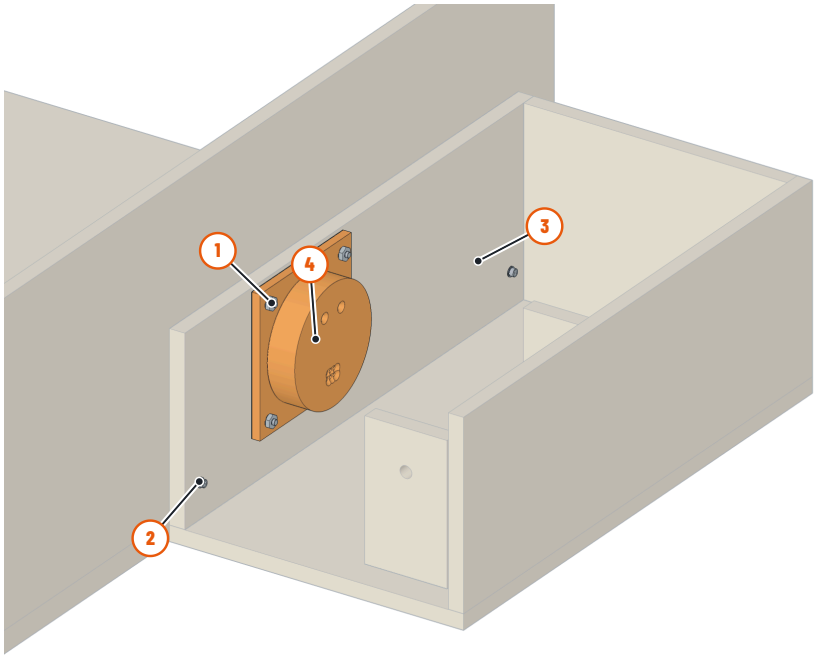
YOU NEED

P18	Box conduit fitting	1
S04	PTFE: stick_hi, stick_lo, bucket_hi, bucket_lo	4

WHY THIS ORDER. The tubes stop in P18 on its sandbox side, and P18 is clamped against the box front by the rod nuts. Once those nuts are on, the window cannot be reached. DERIVED

1. Take the four temporary nuts off the rods outside the sandbox wall (the silicone must have cured).
2. Seat the box end of the four arm tubes in P18's rounded window from the **socket side** (the side with the big round recess). They sit side by side and stop on the window floor.
3. Stand the box a little way from the sandbox (top off) and hold P18 inside it behind the front. Feed the four free tube ends out through the box front's 2-7/16″ hole, through the sandbox wall's hole, across the spacer chamber, through P17's window, into the copper elbow and up the PEX tube until they come out at the top of the turret.
4. Slide the box up to the sandbox wall: the four rods pass through the box front's 3/16″ holes. Slip P18 over the rod ends inside the box, socket toward the box front, and push it flat against the front while you take up the tube slack at the turret.

STAGE 7 — CONTINUED: NUTS, CORNER BOLTS, SEAL



1. 4 × 8-32 rod: nut on the box fitting
2. 2 × M3 × 30 at the lower corners
3. Flat washers both sides; nuts inside the sandbox
4. Box conduit fitting (P18)

Fig. 24 Box against the sandbox wall: four through-rods and two corner bolts (box top, left side and sandbox walls hidden).

YOU NEED (THIS PAGE)

H13	8-32 nut (from the Stage 4 bag)	4
H05	M3 × 30 socket cap	2
H08	M3 flat washer	4
H07	M3 nut	2

5. Run an 8-32 nut onto each rod end on P18's flange. Tighten the four in turn, a little at a time, holding the nut on the pedestal side with the second wrench (reach in through the pedestal's +u hatch). The stack closes up: P18, box front, sandbox wall, spacers, pedestal wall, P17.
6. Fit the two lower-corner bolts: M3 × 30 through the box front and the sandbox wall, flat washers both sides, nut inside the sandbox.
7. At the turret, pull the four tubes up gently and leave them hanging out of the top for Stage 8. Their spare length bows inside the spacer chamber.

✓ **CHECK.** Each tube slides a little when pushed at the turret end and does not come out of P18's window when pulled. The box stands flat on the same table as the sandbox; the pedestal does not move when pushed. VERIFIED

! **DO NOT OVER-TIGHTEN THE RODS.** The nuts bear on printed flanges. Snug plus a quarter turn is enough; crushing the OSB or cracking a flange breaks the seal. ASSUMPTION

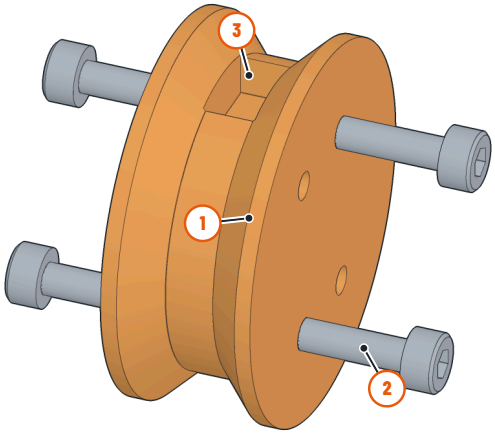
★ **SAND SEAL.** If you see daylight round the outer spacer from inside the sandbox, run a second bead of silicone into the corner between the spacer and the wall.

STAGE 8

ASSEMBLE THE ARM

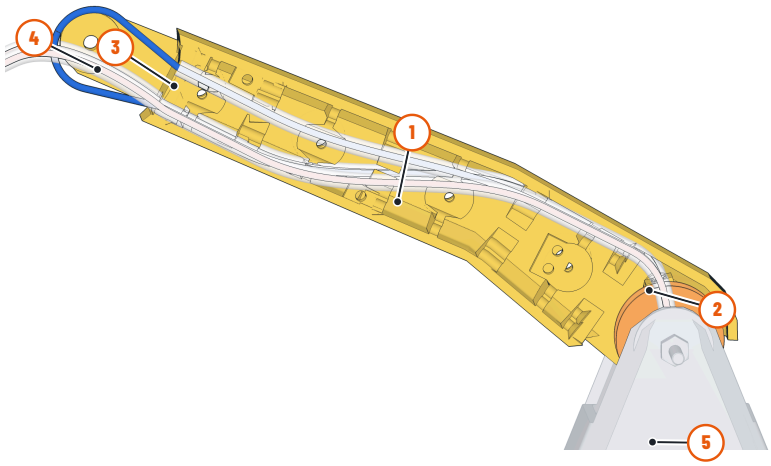
Boom, stick and bucket, each with its drum, its rope's joint end and its PTFE tubes laid in before the halves close.

READ APPENDIX C ITEMS 1-3 FIRST. The CAD shows three fit problems in this stage: the stick-drum key screws collide, two stick lug holes merge, and the joining-lug screws are enclosed once a member is closed. The steps below say where each one matters.



1. Boom drum (P05)
2. M3 × 12 key screw (4: two per face)
3. Crimp pocket (rope midpoint)

Fig. 25 Boom drum with its four key screws.



1. Boom half, right (P02), split face up
2. Boom drum: key-screw heads drop into the wall pockets
3. Stick tubes: to their stop in the boom tip
4. Bucket tubes: on through the boom tip
5. Boom ropes: straight down into the tower

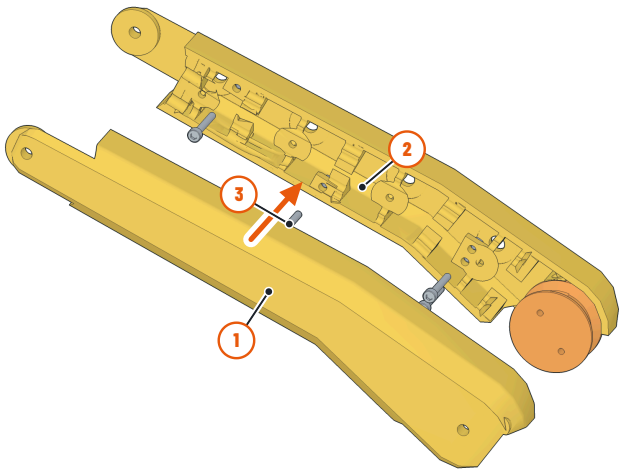
Fig. 26 Boom right half with drum, PTFE tubes and boom rope.

YOU NEED — BOOM

P05	Boom drum	1
H02	M3 × 12 socket cap (key screws)	4
S01	Boom rope 49-13/16″ (midpoint marked)	1
S02	Single crimp sleeve	1
P02	Boom half, right	1
P01	Boom half, left	1
H03	M3 × 20 socket cap (lugs)	4
H07	M3 nut	4

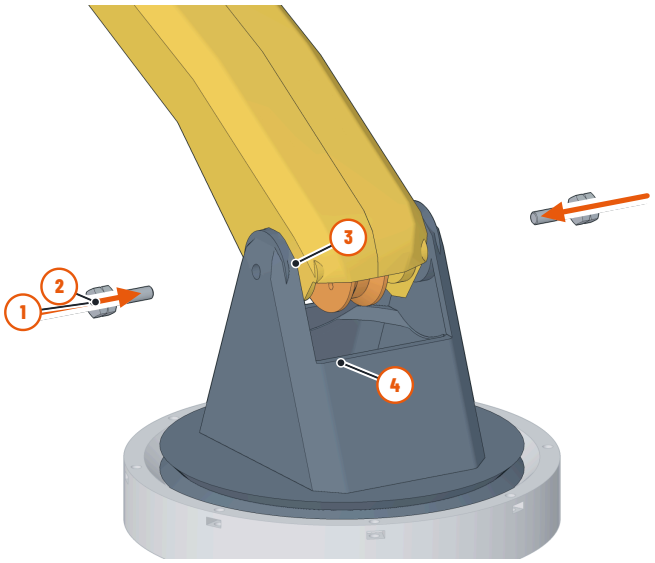
1. Screw two M3 × 12 into each face of the boom drum (4 screws, into the Ø2.6 pilots) until the underside of each head is **10 mm** from the drum face. The heads will sit in pockets in the boom walls. VERIFIED
2. Crimp the single sleeve onto the boom rope at its midpoint mark. Seat the sleeve in the drum's crimp pocket and wrap one tail each way round the groove.
3. Lay the drum in the right boom half (P02) so its two right-hand screw heads drop into the wall pockets at the boom root.
4. Lay the four PTFE tubes from the turret into P02's channels: the two **stick** tubes to their stop at the boom tip; the two **bucket** tubes on through the boom tip. Trim each stick tube so it seats on its stop.

STAGE 8 — CONTINUED: CLOSE THE BOOM AND PIN IT TO THE TOWER



1. Boom half, left (P01)
2. Boom half, right (P02)
3. Joining lugs: M3 × 20 + nut (see Open item 3)

Fig. 27 Closing the boom.



1. 8-32 rod, 1-5/8″ (one per side)
2. Jam pair: 2 × 8-32 nut, outside the tower cheek
3. Inner end flush with the boom's inner wall
4. Tower cheek

Fig. 28 Boom split pins: one short rod per side.

5. Feed both boom rope tails down into the tower and the PEX tube, and push them on down to the box (see the tip).
6. Close the boom: put the left half (P01) on, the drum's left screw heads in its pockets.
7. Joining lugs: 4 × M3 × 20 + nut clamp the halves. Open item 3: in the CAD the lug screws cannot be reached once the halves are closed — decide how you will hold the halves before this step.
8. Set the boom root between the tower cheeks. Push a 1-5/8″ 8-32 rod through each cheek into the boom wall until its inner end is flush with the boom's inner wall.
9. Fit two 8-32 nuts on the outer end of each pin and lock them against each other (jam pair) against the cheek.

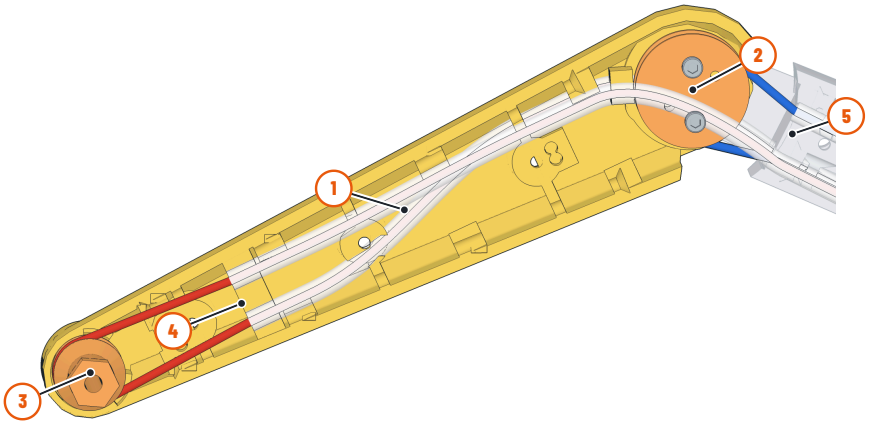
YOU NEED — BOOM PINS

H10	8-32 rod, 1-5/8″	2
H13	8-32 nut	4

★ **THREADING A BARE TAIL DOWN THE ELBOW.** Both fittings have bell-mouth entries on the chamber side. If a boom tail will not find its Ø2.4 hole at the bottom of the elbow, thread it up from the box side instead (through P18, the spacer chamber, P17, the elbow and the PEX), take it over the drum and send the other tail down.

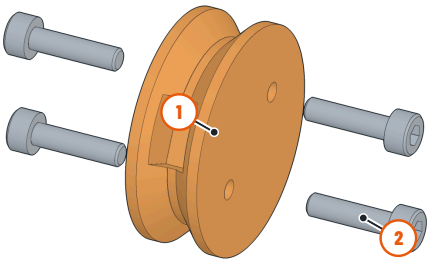
✕ **PIN TOO LONG INSIDE.** A split pin pushed past the boom's inner wall blocks the PTFE tubes that cross the joint axis. The inner ends must be flush. VERIFIED

STAGE 8 — CONTINUED: STICK



1. Stick half, right (P04)
2. Stick drum (P06) on its key screws
3. Bucket drum-axle (P07): journals in the nose holes
4. Bucket tubes: stop in the stick nose
5. Stick tube stops (boom tip)

Fig. 29 Stick right half: drum, bucket drum-axle, tubes and ropes.

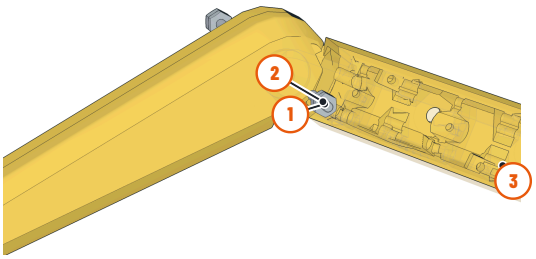


1. Stick drum (P06)
2. Key screws enter the same through-pilot from both sides

Fig. 30 Stick-drum key screws.

YOU NEED — STICK

P06	Stick drum	1
H02	Key screws — see Open item 1	4
S01	Stick rope 67″	1
S02	Single crimp sleeve	2
S01	Bucket rope 80-7/8″	1
P07	Bucket drum-axle	1
P04	Stick half, right	1
P03	Stick half, left	1
H03	M3 × 20 + nut (lugs)	3–4
H11	8-32 rod, 7/8″	2
H13	8-32 nut	4

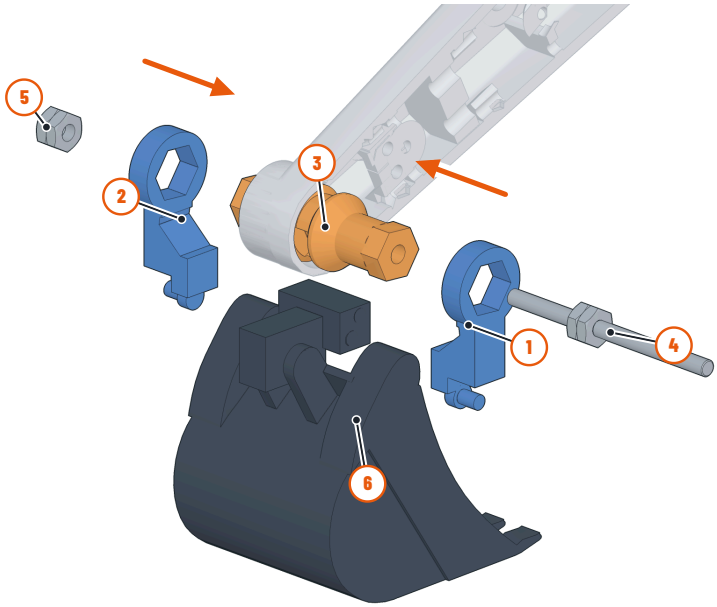


1. 8-32 rod, 7/8″ (one per side)
2. Jam pair outside the boom tip
3. Boom tip

Fig. 31 Stick split pins.

1. Stick drum key screws: each pilot goes right through the 10 mm drum, and the model puts an M3 × 12 into it from each side with the heads 6.2 mm from the faces — the tips then overlap by 1.6 mm. Use M3 × 8 (in your kit) or shorten the M3 × 12 by 2 mm. OPEN ITEM
2. Crimp the stick rope's single sleeve at its midpoint, seat it in the stick drum's pocket and wrap the tails. Lay the drum in P04 with its screw heads in the pockets.
3. Push each stick tail into the end of its stick tube at the boom tip and on down the tube to the box.
4. Bucket drum-axle: put its journals in the nose holes of P04. Crimp the bucket rope's sleeve at its midpoint, seat it in the drum-axle's pocket, wrap the tails, and push them into the bucket tubes (laid in P04's channels to their stop in the nose) and on to the box.
5. Close with P03. Lugs: the two at the stick root overlap (2.9 mm apart) — only one screw fits there (Open item 2).
6. Fit the stick to the boom tip with a 7/8″ 8-32 rod each side (inner ends flush) and a jam pair outside.

STAGE 8 — CONTINUED: BUCKET



1. Ear, left (P09): hex socket on the axle's hex end
2. Ear, right (P10)
3. Bucket drum-axle (P07) in the stick nose
4. 8-32 rod, 2-9/16″: the bucket pin
5. Jam pair each end
6. Bucket (P08): glue faces on the lugs

Fig. 32 Bucket, ears and bucket pin, exploded along the pin.

YOU NEED — BUCKET

P08	Bucket	1
P09	Bucket ear, left (2026-09-29 file)	1
P10	Bucket ear, right (2026-09-29 file)	1
H12	8-32 rod, 2-9/16″ (bucket pin)	1
H13	8-32 nut	4
	CA or epoxy	

! **GLUE LAST.** Glue the ears only with the bucket pin fitted and tightened, so the hex sockets, the pin and the bucket lugs line up.

✕ **EARS SWAPPED.** The left ear (P09) goes on the +v end of the drum-axle. Swapped ears do not seat on the bucket lugs.

1. Dry fit: slide each ear's hex socket onto the drum-axle's hex end (left ear on the +v end). The ears' lower blocks rest on the bucket's lugs and the small pin under each ear goes into the bucket.
2. Push the 2-9/16″ rod through the left ear, the drum-axle's bore and the right ear. Centre it.
3. Fit a jam pair on each end, snug against the ears: the rod clamps the ears to the drum-axle's shoulders.
4. When the bucket hangs straight and swings freely, glue each ear's lower block to its bucket lug (CA or epoxy), as on the original EX-MA. Let it cure fully before rigging.
5. Swing the bucket, stick and boom by hand through their ranges: nothing may rub.

✓ **CHECK — JOINT RANGES.** Boom -40° … +45°, stick -45° … +50°, bucket -55° … +85° from the modelled pose, with no rubbing. The model is contact-free well beyond these (boom −42…+80°, stick −54…+150°, bucket −56…+170°). VERIFIED

STAGE 9

RIG THE ARM ROPES AT THE LEVERS

Set every joint to the middle of its range and every lever upright, then loop each tail onto its anchor and tension it.

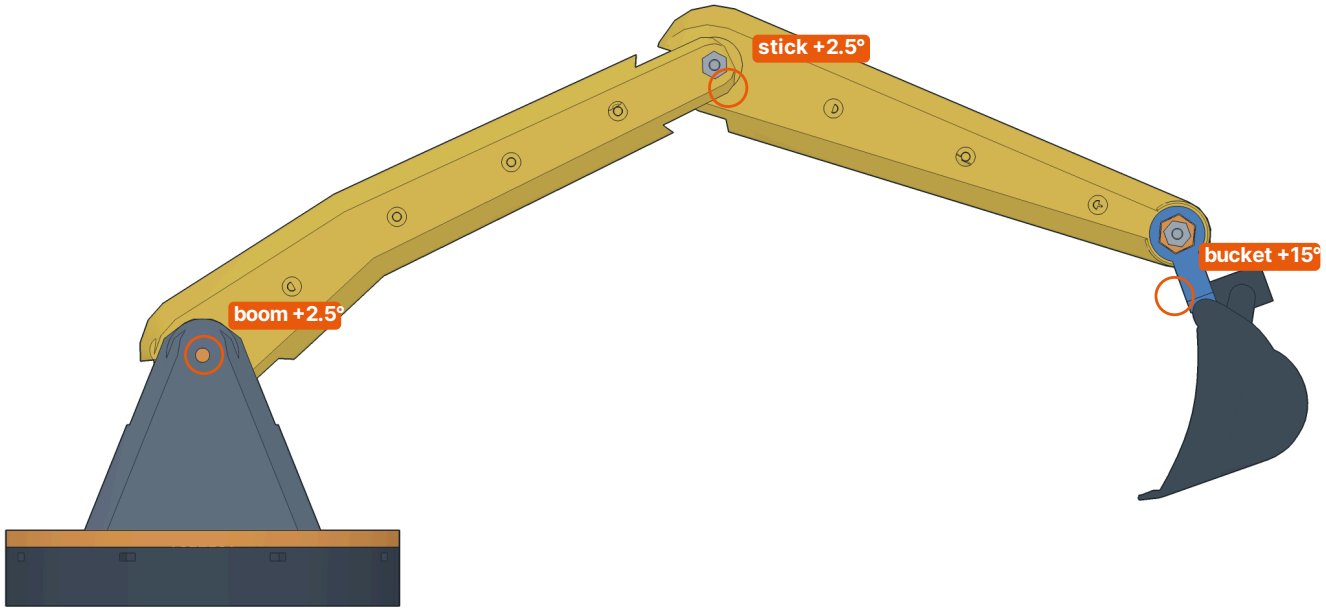


Fig. 33 The rigging pose: each joint at the middle of its working range, measured from the modelled pose (+ = raised).

CONFIGURATION: ZERO POSITIONS VERIFIED

Joint	Working range	Rig at
Boom	−40° … +45°	+2.5°
Stick	−45° … +50°	+2.5°
Bucket	−55° … +85°	+15°
All levers	slot end to slot end	upright
Slew	−90° … +90°	arm straight out, wheel centred

Crimping at mid-range leaves the same rope wrap on the joint drum at both ends of travel. Prop the arm in this pose with a block while you work. VERIFIED

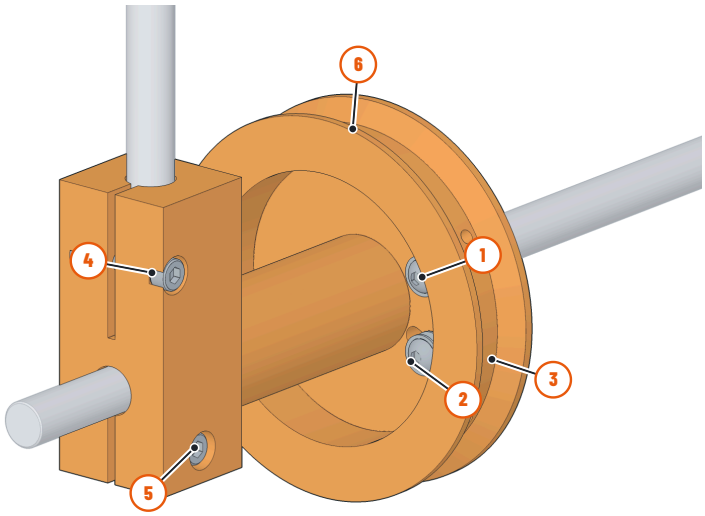
YOU NEED

S03	Double crimp sleeve	6
H01	M3 × 10 button head (in the hubs)	6
H08	M3 flat washer	12
H09	M3 split-lock washer	6
H07	M3 nut	6

1. Inside the box each tail comes out of P18's box face. Take each strand to its own lever drum: boom tails to the boom hub, stick to the stick hub, bucket to the bucket hub. Strands never cross.
2. Wrap the tail into the groove and pass it through the rim hole beside its anchor slot (upper strand to the upper anchor, lower to the lower).
3. With the anchor screw at the **outer** end of its slot (nearest the rope hole), pull the rope just taut and crimp a loop with a double sleeve so that it sits on the screw.
4. Stack on the screw: flat washer — rope loop — flat washer; the split-lock washer and nut on the other side of the web. Snug only.

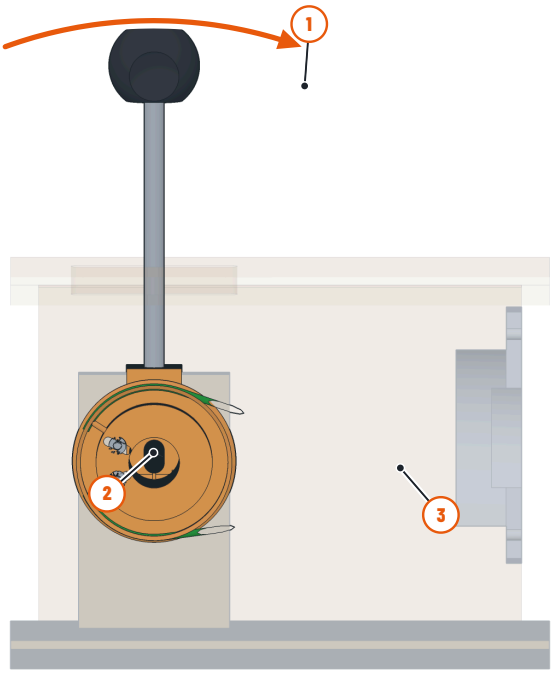
> **CRIMPING WITH THE JOINT OFF-CENTRE.** A crimp made with the joint at one end of its range leaves no rope wrap at the other end, and the lever stops short. Re-check the rigging pose before every crimp.

STAGE 9 — CONTINUED: TENSION AND DIRECTION



1. Rope anchors: M3 × 10 button head in the slot (tension adjuster)
2. Washer — rope loop — washer under the head
3. Rope holes through the rim into the groove
4. Handle socket + pinch clamp (M3 × 20)
5. Drag clamp (M3 × 20 + lock washer)
6. Ø60 drum groove

Fig. 34 Lever hub (boom): anchors, rope holes and clamps.



1. Push = forward (toward the sandbox)
2. Lever axle (5/16" rod)
3. Boom rope strands to the box fitting

Fig. 35 Boom lever from the side (box cut away): push = forward.

5. Tension: slide each anchor screw along its slot **toward the drum centre** (away from the rope hole) until both strands are taut with the joint still at mid-range and the lever upright, then tighten the nut. Each slot gives about 8 mm of take-up.
6. Repeat for the other strand of the same circuit so the lever stays upright when both are taut.
7. Direction: push the lever forward (toward the sandbox). It must give boom down, stick out, bucket dump. If one is reversed, undo that rope at the joint drum and wrap it the other way round the drum.
8. Move each lever slowly end to end and watch its joint: smooth, no slack strand, the rope stays in every groove.

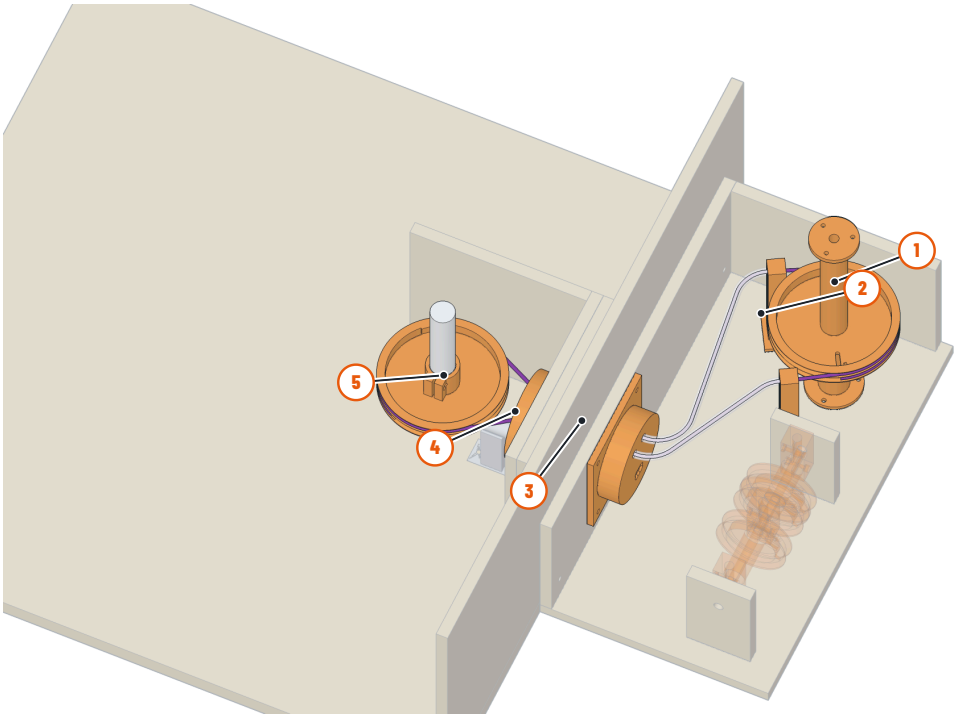
? **OPEN ITEM 6 — WHICH WAY TO SLIDE.** The checklist says "slide the anchors outward". In the CAD the rope enters each slot at its **outer** end, so moving the screw inward is what takes up rope. Trust what you see: the strand tightens as the screw moves away from the rope hole. VERIFIED

✓ **CHECK — NO COUPLING.** Hold the bucket lever still and move the boom and stick levers through their full travel: the bucket must not move. The model predicts at most 0.66 mm of rope movement from tube bending ("Bowden coupling"). VERIFIED

STAGE 10

RIG THE SLEW ROPES

Two ropes in two PTFE sleeves: each one ends in a pocket on the pedestal drum and in a loop on the spool.



1. Slew spool (wheel)
2. Sleeve posts: sleeve ends seat here
3. PTFE sleeves through the box fitting and the spacer chamber
4. Pedestal fitting: angled counterbores (sleeve stops)
5. Slew drum: crimp pockets

Fig. 36 Slew path from above (box, pedestal and sandbox walls cut away): spool → posts → sleeves → spacer chamber → pedestal fitting → slew drum.

YOU NEED

S04	PTFE slew sleeve A 10-15/16″, B 13-5/8″	2
S01	Slew rope A 30-5/16″, B 33-1/16″	2
S02	Single crimp sleeve	2
S03	Double crimp sleeve	2
H01	M3 × 10 button head (in the spool)	2
H08	M3 flat washer	4
H09	M3 split-lock washer	2
H07	M3 nut	2

1. Push sleeve A from its post (the one nearer the levers) through P18's upper slew hole on that side, across the spacer chamber and into P17's angled counterbore until it stops. Seat its other end in post A's counterbore and trim the 3/8″ extra. Same for sleeve B.
2. Crimp a single sleeve on one end of rope A. Thread the other end through the pedestal-fitting exit, sleeve A and post A to the spool.
3. Seat rope A's crimp in the **upper**-lane pocket of the pedestal drum and wrap it half a turn in that lane. Rope B: lower lane.

✕ **SWAPPED LANES.** Rope A (the shorter, 30-5/16″) runs in the **upper** lane on both drums and in sleeve A (10-15/16″). If ropes cross between the drums, the arm turns the wrong way.

STAGE 10 — CONTINUED: DRUM POSITION, SPOOL LOOPS, TENSION

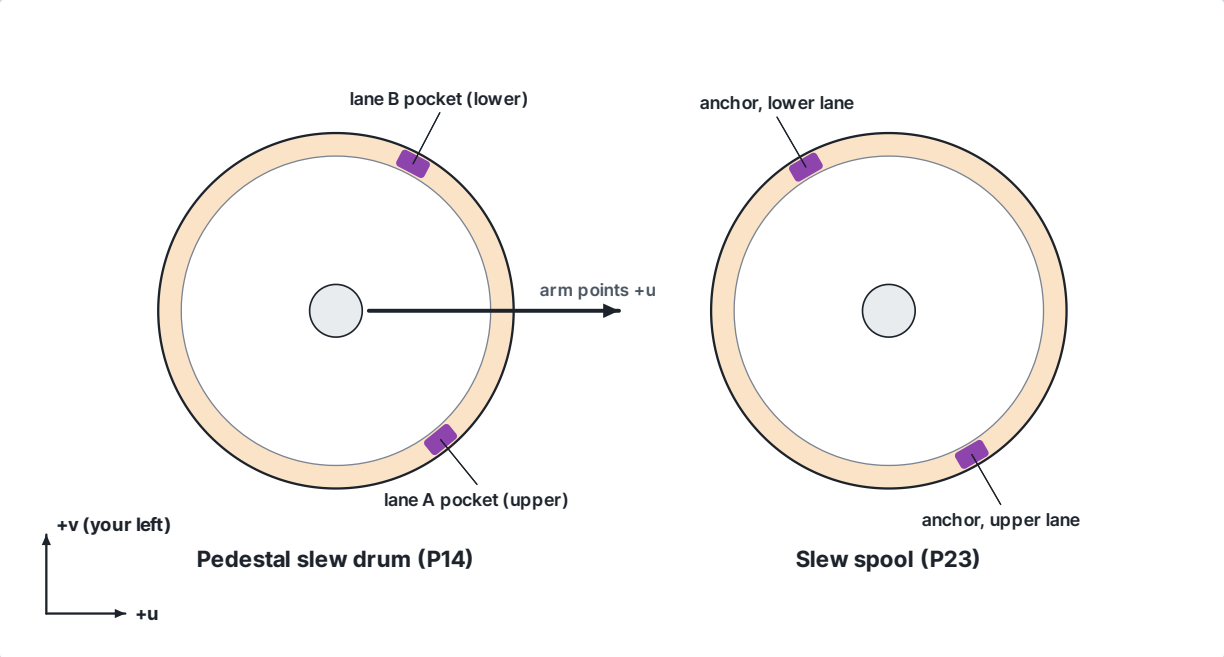
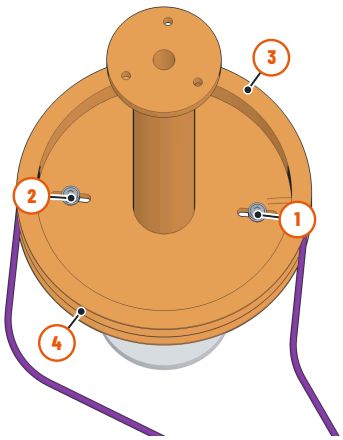


Fig. 37 Zero position seen from above, arm pointing straight out (+u): pocket and anchor angles from the model. VERIFIED

4. Point the arm straight out (+u, away from the box). Turn the pedestal drum on the PEX tube until its pockets are at the angles in the diagram (lane A 51° to the right of the arm, lane B 62° to the left), then tighten its pinch clamp hard. The drum must not slip on the tube. VERIFIED
5. Turn the spool so its two anchors are at the angles in the diagram (wheel centred).
6. Wrap each rope half a turn round its spool lane, pass it through the rim hole by its anchor, and crimp a loop (double sleeve) onto the anchor screw with the screw at the outer end of the slot: washer — loop — washer, lock washer and nut below the web.
7. Tension both ropes by sliding the anchors toward the spool centre, keeping the arm straight out. Each slot gives about 12 mm.
8. Turn the wheel left: the arm must swing to your left, 1 : 1, up to 90°. Then right to 90°.



1. Anchor A: M3 × 10 button head, washers each side of the loop
2. Anchor B
3. Upper lane
4. Lower lane

Fig. 38 Spool anchors and lanes from above.

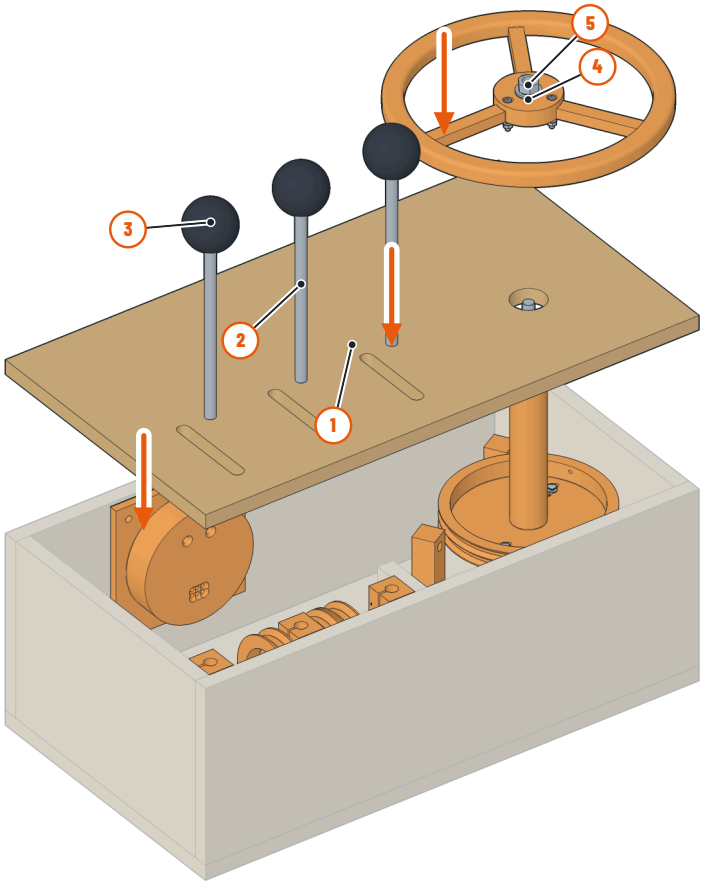
✓ **CHECK — WRAP LEFT AT THE ENDS.** At ±90° each rope must still have at least 90° of wrap on both drums (the model has 90–270°). If a rope runs out early, the drum or the spool was not centred: re-centre and re-crimp. VERIFIED

! **DRUM CLAMP.** A slipping pedestal drum feels like "the wheel turns but the arm doesn't". Tighten the M3 × 25 pinch clamp before tensioning, and re-check it after the first test.

STAGE 11

TOP PANEL, HANDLES, KNOBS AND WHEEL

The top panel's slots are the lever end stops. Fit it, push the handles through, and bolt the wheel on, 3-3/4" above the lid.



YOU NEED

W06	Box top (3 slots, 1 hole)	1
H20	#6 × 1-1/4" wood screw	16
H16	5/16" rod handle, 5-11/16"	3
P22	Lever knob	3
H03	M3 × 20 (handle clamps)	3
H03	M3 × 20 (drag clamps)	3
H09	M3 split-lock washer	9
H07	M3 nut	9
P24	Slew wheel	1
H03	M3 × 20 (wheel)	3
H17	5/16" nut + washer (spool top)	1
	CA or epoxy	

1. Top panel (W06)
2. 5/16" rod handle, 5-11/16" (×3)
3. Lever knob (P22 ×3), glued
4. Slew wheel (P24): 3 × M3 × 20 + lock washer + nut
5. 5/16" washer + nut on the spool axle

Fig. 39 Top panel, handles, knobs and wheel.

1. Lower the top onto the box: the spool tube passes through its 1" hole (the top is the spool's upper bearing). Screw it to the front, rear and both sides, 4 screws per edge — screws only, no glue, so it lifts off for service.
2. Push each handle rod down through its slot into its hub socket (25 mm deep). Tighten the handle pinch clamp (M3 × 20, lock washer and nut).
3. Glue a knob on each handle: the rod goes 14 mm into the knob's bore.
4. Bolt the wheel to the spool's top flange: 3 × M3 × 20, heads in the wheel's counterbores, lock washer and nut under the flange.
5. Fit the washer and nut on top of the spool axle. Snug it until the spool has no up-down play but still turns freely. ASSUMPTION
6. Fit the three drag-clamp screws (M3 × 20, lock washer, nut) loosely; they are set in Stage 12.

✓ **CHECK.** Each lever stops on the end of its slot at both ends of travel — the slot is the hard stop. The model has the handle touch the slot end exactly at the end of travel, with nothing else touching. The bucket knob passes 5/8" from the wheel rim at its end of travel (Open item 6). VERIFIED

STAGE 12

FINAL TEST AND CONFIGURATION

Run every control through its full range, check for coupling, then set the lever friction.

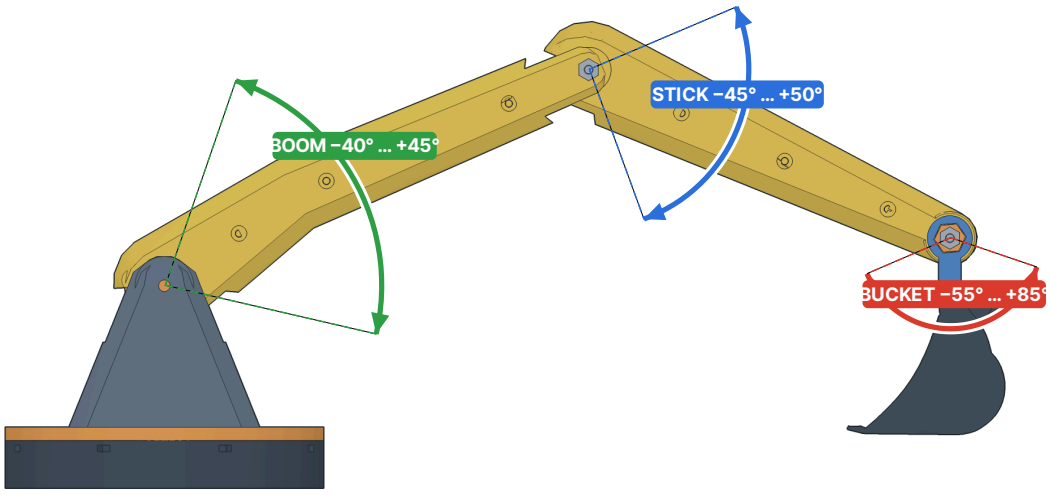


Fig. 40 Working ranges set by the lever slots, measured from the modelled pose shown (+ = raised).

FUNCTION TEST VERIFIED

1. Each lever, slowly, end to end: its joint moves through its whole working range and the handle stops on the slot ends.
2. Wheel to −90° and +90°: the arm follows 1 : 1 and no rope jumps its groove.
3. No coupling: move the boom and the stick through their ranges while watching the bucket. It must not move.
4. At every end stop, look at the ropes: each stays in its groove, and no PTFE tube is pinched at a joint.
5. Turn the turret by the wheel through ±90°: smooth, no knocking from the ball ring.

CONFIGURATION: LEVER HOLDING FRICTION

6. Drag clamps: with the arm extended, tighten each lever's drag-clamp screw a little at a time until the lever holds its position when you let go. The boom needs the most; the bucket almost none.
7. Re-tension every circuit after the first hour of use (rope stretch and crimp seating), using the slotted anchors.

Lever	Holding torque (model)	Clamp force (μ 0.3)
Boom	0.65 N·m	172 N
Stick	0.23 N·m	60 N
Bucket	0.04 N·m	10 N

From the box validation report VERIFIED. Use them to compare levers: the boom clamp needs several times the stick's.

✓ **DONE.** When every check on this page passes, the build is complete. Keep the hardware bags: the spares are listed in Appendix A.

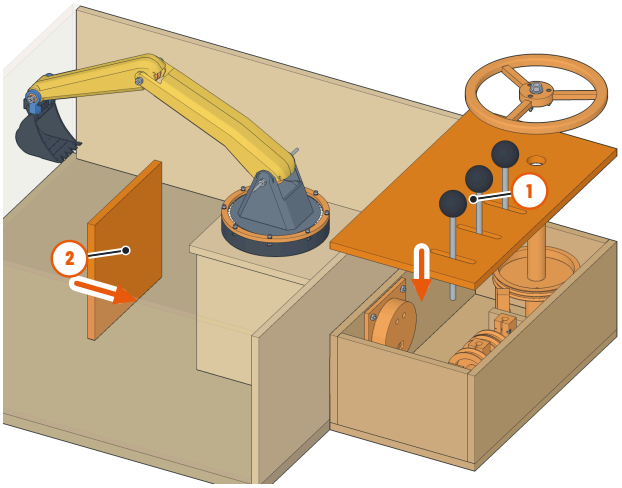
AFTER ASSEMBLY

OPERATION AND MAINTENANCE

OPERATING

- Stand behind the box. Left hand: boom and stick levers. Right hand: bucket lever and slew wheel.
- Move one control at a time at first. Push forward = boom down, stick out, bucket dump; wheel left = arm left.
- The slot ends are the stops. Do not lever against them or force the wheel past ±90°.
- Do not lift or carry the machine by the arm, the levers or the wheel. Carry the sandbox and the box together from below.
- Dig in dry sand only. Brush sand off the slewing ring and out of the lever slots after use.

SERVICE ACCESS VERIFIED



1. Box top: lift off (screws only)
2. Pedestal wall +u: unscrew (move the sand away first)

Fig. 41 The two service hatches: box top and pedestal wall +u.

- **Box top** — lift off (screws only): lever hubs, anchors, spool, sleeve posts, box fitting.
- **Pedestal wall +u** — unscrew (screws only): slew drum clamp, copper elbow, sleeve ends, the through-rod nuts. Clear the sand from that side first.
- **An arm half** — opening one half gives access to its drum, crimp pocket and tubes. See Open item 3 for how the halves are held.

MAINTENANCE SCHEDULE ASSUMPTION

When	Do
After the first hour	Re-tension all four circuits (rope stretch and crimp seating). VERIFIED
Every session	Brush off sand; check the knobs and the handle clamps are tight.
Monthly	Check the jam nuts on the five joint pins and the 8 retaining-ring screws; check the slew drum clamp.
If a lever feels slack	Re-tension at its anchors (Stage 9).
If the arm drifts	Tighten that lever's drag clamp (Stage 12).

REPLACING A ROPE DERIVED

1. Lift the box top; undo the two loops of that circuit at its lever drum.
2. Open the arm half that holds its joint drum and lift the crimp out of the pocket.
3. Pull the old rope out from the arm end; use it to measure the new one (cut lengths in Appendix B).
4. Re-rig as in Stages 8 and 9 (or 10 for slew).

AFTER ASSEMBLY

TROUBLESHOOTING

Find the symptom, check the likely cause, then go to the stage named in the fix. Causes come from how the circuits are built; where a cause is general practice it is not tagged separately.

Symptom	Likely cause	Fix
A lever moves the wrong joint	Strands crossed in the box, or ropes swapped at the box fitting	Each circuit's two strands go to its own hub; the box fitting columns run boom / stick / bucket in the same order as the hubs (Stage 9).
A joint moves the wrong way	Rope wrapped the wrong way round the joint drum	Reverse that rope's wrap on its joint drum (Stage 9, step 7).
Moving one lever moves another joint	PTFE tube not seated at the box fitting or at its arm stop; tube pinched	Re-seat both tube ends; check the tube is not trapped by a split pin or lug (Stage 8).
Lever stops short of the slot end	Rope crimped with the joint off-centre; rope too tight	Return to the rigging pose and re-crimp (Stage 9).
Lever floppy, joint lags	Rope slack	Slide the anchors toward the drum centre (Stage 9, step 5).
Arm sags when you let go	Drag clamp too loose	Tighten that lever's drag clamp (Stage 12, step 6).
Lever very stiff	Drag clamp too tight; axle nuts squeezing the hubs; hub rubbing a neighbour	Ease the drag clamp; back the axle nuts off; hubs need their 0.5 mm gaps.
Rope jumps out of a groove	Slack rope at an end stop	Re-tension; check the joint is not driven past its working range.
Wheel turns, arm does not (or slips)	Pedestal slew drum or turret-hub clamp slipping on the PEX tube	Tighten both M3 × 25 pinch clamps (Stages 5 and 10).
Arm turns the opposite way to the wheel	Slew ropes crossed between lanes	Rope A upper lane on both drums, rope B lower (Stage 10).
Slew range less than ±90°	Drum or spool not centred when the ropes were crimped	Re-centre both (Stage 10, steps 4–5) and re-crimp.
Turret rocks or lifts	Retaining ring too high	Shim under the retaining ring (Stage 5 check).
Turret binds or knocks	Missing or extra BB; uneven ring screws; sand in the race	57 balls; loosen the ring screws evenly; clean the race.
Bucket wobbles on its pin	Jam nuts loose; ear glue joint failed	Re-lock the jam pairs; re-glue the ear with the pin fitted.
Handle turns in its hub	Handle pinch clamp loose	Tighten the M3 × 20 handle clamp (Stage 11).
Sand coming out at the box	Spacer seal leaking at the sandbox wall	Run silicone into the corner between the outer spacer and the wall from inside the sandbox (Stage 7 tip).
Rope frays at a fitting	Burr on the copper end or a fitting hole	Deburr; check the rope runs through the bell-mouth holes.

APPENDIX A

BILL OF MATERIALS

From `2026-09-24-bill-of-materials.md` VERIFIED; updated for the minimal build (no conduit, 8-32 through-rods, kept prints).

PRINTED PARTS

ID	Part	Qty	Material	Plate
P01	Boom half, left (+v)	1	PETG	arm 1
P02	Boom half, right (−v)	1	PETG	arm 1
P03	Stick half, left (+v)	1	PETG	arm 1
P04	Stick half, right (−v)	1	PETG	arm 1
P05	Boom drum (Ø30 pitch)	1	PLA+	other 6
P06	Stick drum (Ø26 pitch)	1	PLA+	other 6
P07	Bucket drum-axle (Ø17 pitch)	1	PLA+	other 6
P08	Bucket	1	PLA+	other 6
P09	Bucket ear, left (G)	1	PLA+	other 6 (new file)
P10	Bucket ear, right (H)	1	PLA+	other 6 (new file)
P11	Tower (turret)	1	PLA+	other 4
P12	Base (lower race)	1	PLA+	other 4 (new file)
P13	Slewing-ring retaining ring	1	PLA+	other 5
P14	Pedestal slew drum (Ø110)	1	PLA+	other 1
P15	PEX bushing	1	PLA+	other 1
P16	Elbow support	1	PLA+	other 1
P17	Conduit fitting, pedestal end	1	PLA+	other 1
P18	Conduit fitting, box end	1	PLA+	other 3
P19	Lever hub, boom	1	PLA+	other 2
P20	Lever hub, stick	1	PLA+	other 2
P21	Lever hub, bucket	1	PLA+	other 3
P22	Lever knob (Ø36)	3	PLA+	other 2
P23	Slew spool (Ø110)	1	PLA+	other 3
P24	Slew wheel (Ø184)	1	PLA+	other 2
P25	Spool riser	1	PLA+	other 2
P26	Slew sleeve post	2	PLA+	other 2

HARDWARE

ID	Item	Qty	Used for
H01	M3 × 10 button head	8	rope-tail anchors on the 3 lever hubs (2 each) and the slew spool (2)
H02	M3 × 12 socket head cap	16	boom + stick drum key screws (8), retaining ring (8)
H03	M3 × 20 socket head cap	17	joining lugs (8), drag clamps (3), handle clamps (3), slew wheel (3)
H04	M3 × 25 socket head cap	2	PEX pinch clamps: turret hub (1), slew drum (1)
H05	M3 × 30 socket head cap	2	box front to sandbox wall, lower corners
H06	8-32 threaded rod, 3-1/16″	4	through-rods: box fitting → pedestal fitting (replace the 8 M3 fitting bolts)
H07	M3 hex nut	37	with the M3 screws above
H08	M3 flat washer	20	corner bolts (4), rope anchors (16)
H09	M3 split-lock washer	27	under nuts/heads on moving or clamping joints
H10	8-32 rod, 1-5/8″ (42 mm)	2	boom split pins
H11	8-32 rod, 7/8″ (22 mm)	2	stick split pins
H12	8-32 rod, 2-9/16″ (65 mm)	1	bucket pin
H13	8-32 hex nut	20	jam pairs on the pins (12), through-rods (8)
H14	5/16″ rod, 8-1/16″	1	lever axle
H15	5/16″ rod, 10-13/16″	1	slew spool axle
H16	5/16″ rod, 5-11/16″	3	lever handles
H17	5/16″ nut + SAE washer	4	2 on the lever axle, 2 on the spool axle
H18	#8 × 1″ flat-head wood screw	4	base to the pedestal top
H19	#6 × 3/4″ flat-head wood screw	14	bearing blocks (4), riser (3), posts (4), PEX bushing (3)
H20	#6 × 1-1/4″ flat-head wood screw	121	panel joints, 4 per edge, with wood glue (120); elbow support (1)

M3 counts all fit in the Fgruh kit. Spares left: about 15″ of 8-32 rod and 2 nuts; #4-40 and #6-32 screws unused.

APPENDIX B

ROPE, TUBE AND ROD REFERENCE

CUT LENGTHS DERIVED

Piece	Cut	mm	Modelled path (mm)
Boom rope	**49-13/16″**	1265	1145 + 2 × 60 loops
Stick rope	**67″**	1702	1582 + 2 × 60
Bucket rope	**80-7/8″**	2054	1934 + 2 × 60
Slew rope A	**30-5/16″**	770	710 + 60
Slew rope B	**33-1/16″**	840	780 + 60
PTFE stick_hi	**23-1/8″**	587	566 + 21 spare and trim
PTFE stick_lo	**23-1/2″**	598	576 + 21 spare and trim
PTFE bucket_hi	**29-5/8″**	753	732 + 21 spare and trim
PTFE bucket_lo	**30″**	762	741 + 21 spare and trim
PTFE slew sleeve A	**10-15/16″**	278	256 + 22 spare and trim
PTFE slew sleeve B	**13-5/8″**	346	324 + 22 spare and trim

Paths from `cable-paths.py` on the minimal model. Each tube keeps the same allowance over its path as in the full design: 3/8″ to trim plus the spare length that bows in the chamber.

THE STACK, BOX TO PEDESTAL DERIVED

Layer	Thickness
P18 flange (box fitting)	6 mm
Box front W02	7/16″
Sandbox wall W18	7/16″
Spacer frames W22 + W23	7/8″
Pedestal wall W10	7/16″
P17 flange (pedestal fitting)	6 mm
Rod: stack + a nut and 1/16″ of thread each end	**3-1/16″**

Chamber between the fitting floors: 3-1/8″; spare-tube bow radius about 1-3/16″ (5/8″ minimum).

ROPE TRAVEL PER CIRCUIT VERIFIED

Circuit	Range	Rope travel per side	Control
Boom	85°	22.3 mm	lever 42°
Stick	95°	21.6 mm	lever 41°
Bucket	140°	20.8 mm	lever 40°
Slew	180°	172.8 mm	wheel 180°

SLEW WRAP VERIFIED

Slew	Rope A spool / pedestal	Rope B spool / pedestal
−90°	90° / 270°	270° / 90°
0°	180° / 180°	180° / 180°
+90°	270° / 90°	90° / 270°

APPENDIX C

OPEN ITEMS AND VERIFICATION

Found by checking the project's CAD and hardware model. Items 1–6 and 8 are in the project files as they are; item 7 comes from the minimal build with your prints. Each needs your decision. OPEN ITEM

#	What the files show	Options
1	**Stick-drum key screws collide** The stick drum is 10 mm thick and each key pilot runs right through it. The hardware model puts an M3 × 12 into each pilot from each side; the two tips overlap by 1.6 mm.	Use M3 × 8 socket caps on the stick drum (in your kit), or grind 2 mm off each M3 × 12.
2	**Stick root lugs overlap** Joining lugs 1 and 2 of the stick are 2.9 mm apart, so their holes merge and their screws and nuts overlap.	Fit one M3 × 20 in that lug; the stick then has 3 lug screws (1 spare).
3	**Joining-lug screws are enclosed** On both members every lug screw head sits inside the closed shell, and the outer walls have no access hole on the screw axis.	Decide before Stage 8: (a) drill a small access hole on each lug axis; (b) close the halves without lug screws, held with tape or dots of CA along the seam; (c) add access in the CAD and reprint the halves.
4	**5/16″ axle ends** Both axles carry 5/16″ nuts. One design note calls the rod "smooth".	Threaded rod: nothing changes. Smooth rod: thread the ends with a 5/16-18 die or use 5/16″ shaft collars.
5	**Pedestal not screwed to the floor** The cut list says the pedestal screws to the sandbox floor, but no screws are counted. In the minimal build the four rods clamp it to the sandbox wall.	Optional: 4 extra #6 × 1-1/4″ up through the floor into the ±v walls (Stage 4).
6	**Tension direction** The checklist says to slide the rope anchors "outward". In the CAD the rope enters each slot at its outer end, so the strand tightens as the screw moves inward.	Slide toward the drum centre (Stages 9 and 10).
7	**Bucket knob near the slew wheel** With the kept 145 mm spool the wheel sits 3-3/4″ above the lid. At the bucket lever's end of travel its knob comes within 15.5 mm (5/8″) of the wheel rim; the full design had 24 mm.	Works as is; mind your knuckles on the bucket lever. A reprinted short spool (compact spec) would lower the wheel.
8	**Slicer plates hold the old base and ears** Plates 4 and 6 of the PLA+ project were made before the 2026-09-29 rebuilds.	If you have not printed P09, P10 and P12 yet, use the 2026-09-29 STL files (Stage 1).

WHAT WAS VERIFIED FOR THE MINIMAL BUILD, AND HOW

Geometry	A copy of the project sources with the minimal constants (7/16″ OSB, 5/8″ floor, no shim, spacers, compact box with the kept prints) rebuilt every panel, placement and rope path. Illustrations are rendered from it.
Interference	Every pair of parts and fasteners checked against the full design. The only new contact is the elbow support on the floor, which is intended. The left corner bolt was moved 5/16″ left to clear the spacers.
Not verified	Physical fit of real prints, OSB and particle-board screw holding, glue strength, rope friction. Items tagged ASSUMPTION are general practice.

APPENDIX D

DIFFERENCES FROM THE FULL DESIGN

What the minimal build changes, compared with the first manual (2026-10-10). Arm, turret, slewing ring, drums, fittings and every printed part are the same. VERIFIED

	Full design (2026-10-10 manual)	Minimal build (this manual)
Box to pedestal	141 mm of 2″ PVC conduit across an 81 mm gap	no conduit: 2 OSB spacer frames, 3-1/8″ chamber
Fitting fasteners	4 × M3 × 35 (box), 4 × M3 × 25 (pedestal)	4 × 8-32 rod 3-1/16″, nut each end
Wood	12 mm plywood, 21 panels	7/16″ OSB (22 pieces) + 5/8″ particle-board floor
Shim under the box	none needed (12 mm floor)	none: walls 5/32″ shorter instead
Sandbox	610 × 610 × 242 mm	24″ × 24″ × 9-1/2″
Pedestal	200 × 180 × 190 mm	7-3/8″ × 7-1/16″ × 7-5/16″
Control box	250 × 405 × 242 mm	8-7/16″ × 15″ × 6-1/8″
Lever handles	230 mm rod	5-11/16″ rod
Lever slots	133 / 129 / 124 mm	66 / 64 / 62 mm (same swing, shorter lever)
Slew wheel	on the lid	3-3/4″ above the lid (spool kept)
Elbow support	as printed	base plate sawn off; 1 screw from below
Arm ropes	1384 / 1821 / 2173 mm	49-13/16″ / 67″ / 80-7/8″
PTFE + sleeves	3.7 m	130-13/16″
Hardware IDs	H06 = M3 × 35	H06 = 8-32 through-rod; H04 2 pcs; nuts 37; flat washers 20
Printed parts	26 types, 29 prints	the same files, as printed

SAME PRINTED PARTS. P17 and P18 are still called “conduit fittings”: their big sockets now open into the spacer chamber instead of holding the PVC. Nothing is reprinted.

DO NOT MIX THE MANUALS’ CUT SIZES. Every wood size, rope, tube and rod length in this manual is for the minimal build with your boards. The first manual’s millimetre sizes are for 12 mm plywood and the conduit.

APPENDIX E

FILES AND REVISION

PROJECT FILES USED

STL/	printed parts (P01–P26), print-ready
Source/	build scripts; parameters in 2026-09-24-exma_common.py
Validation/	box, kinematics and wall-thickness reports
2026-09-24-bill-of-materials.md	quantities, plates, materials
2026-09-28-compact-control-box-spec.md	compact box depth and height
2026-10-10-assembly-manual/	first manual: renderer, page helpers, styles
2026-10-11-ex-ma-cut-plan-your-boards.pdf	earlier cut plan (compact 14-1/4″ box): superseded by Stage 2 here

REVISION

Date	Change
2026-10-11	First edition of the minimal-build manual.

THIS MANUAL'S SOURCE (2026-10-11-MINIMAL-MANUAL/)

source/2026-10-11-make-minimal-source.py	minimal copy of Source/ (constants + 3 patches)
source/2026-10-11-minimal-scene-export.py	geometry → scene for the renderer
source/2026-10-11-minimal-figure-specs.py	cameras, callouts, exploded offsets
source/2026-10-11-build-minimal-manual.py	text, cut plan and page layout
source/2026-10-11-build-minimal-manual.sh	runs the whole chain
figures/	rendered figures + callout data

The renderer, figure runner, overlay helpers, styles and PDF printer are shared with the first manual (2026-10-10-assembly-manual/source/).